Golf Launch Monitor Technology Review

Measurement Principles, Patent Landscape, and Club/Ball Parameter Estimation

A technical foundation for the OpenFlight open-source launch monitor

OpenFlight Development

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Abstract

This review surveys the measurement technology behind commercial golf launch monitors — Doppler-radar systems (TrackMan, FlightScope, Garmin R10, Full Swing KIT), photometric camera systems (Foresight GCQuad/GC3, Uneekor, SkyTrak, ProTee VX, Garmin R50), and the hybrid radar+camera architectures that now dominate the high end — with emphasis on *how each system measures or infers club-delivery parameters* (club speed, club path, attack angle, face angle, dynamic loft, impact location) *and ball-launch parameters* (ball speed, launch angles, spin rate, spin axis). Primary sources include the governing patent families (TrackMan/Tuxen radar spin and spin-axis patents, the FlightScope/EDH dielectric-lens spin patent, the Wintriss/Foresight photometric patents, and the Acushnet stereo launch-monitor lineage), regulatory (FCC) filings, manufacturer technical documentation, and the peer-reviewed validation literature. The review develops the underlying physics — the D-plane impact model, oblique-impact spin generation, gear effect, and golf-ball aerodynamics — as the common mathematical core every launch monitor implements, and closes with a freedom-to-operate map of the patent landscape and concrete architectural recommendations for OpenFlight, an open-source Doppler-radar launch monitor built on the OmniPreSense OPS243-A and low-cost angle-radar modules. The central finding: every parameter a launch monitor reports sits on a *directness hierarchy* from measured to modeled, the industry has converged on optical augmentation of radar (and radar augmentation of cameras) precisely because neither modality can measure the full parameter set alone, and the recently expired Wintriss photometric patents (2025) open a practical low-cost optical path to true club-face and spin measurement that complements OpenFlight's radar core.

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Chapter 1

Introduction

1.1 Purpose and scope

A golf launch monitor answers two questions for every shot: *what did the club do through impact* (delivery), and *what did the ball do after impact* (launch and flight). Commercial systems answer these questions with strikingly different sensor architectures — continuous-wave Doppler radar, multi-camera photogrammetry, and, increasingly, fusions of the two — yet they all implement the same underlying physics: an oblique-impact collision model connecting club delivery to ball launch, and an aerodynamic model connecting ball launch to flight.

This review is written to support the development of **OpenFlight**[1], an open-source launch monitor. Its goals are:

1. catalogue *how each major commercial system works*, at the level of radar physics, imaging geometry, and signal processing;
2. establish, parameter by parameter, *what is measured directly, what is inferred through a model, and what is essentially estimated* — especially for club-delivery data such as face angle and club path;
3. survey the governing *patent landscape*, since the strongest public documentation of these proprietary methods is found in the patents themselves, and since freedom-to-operate matters to an open-source project;
4. collect the *physics and algorithms* (D-plane, impact mechanics, gear effect, aerodynamic coefficients, spectral spin estimation, stereo triangulation, Kalman filtering) that any implementation needs; and
5. translate all of the above into *concrete design guidance* for OpenFlight's radar-first architecture and its possible optical extensions.

1.2 The two sensing families

Doppler radar systems (TrackMan, FlightScope, Garmin Approach R10, Full Swing KIT) illuminate the hitting area and downrange volume with a microwave carrier and extract target radial velocity from the Doppler shift. Multiple receive antennas turn a velocity sensor into a 3D tracker via phase interferometry (Chapter 4). Radar excels outdoors: it tracks the entire flight, so carry and curvature are *observed*, not modeled. Its weaknesses are at the club face — radar cannot see face orientation directly — and indoors, where a screen truncates the observable flight.

Photometric (camera) systems (Foresight GC-series, Uneekor, SkyTrak, ProTee VX, Garmin R50) capture a burst of high-speed, IR-strobed stereo images over the first ∼30 cm of ball flight

[1] `https://github.com/jewbetcha/openflight` — AGPL-3.0; a DIY launch monitor built on the OmniPreSense OPS243-A 24 GHz Doppler radar, RFbeam K-LD7 angle radars, a sound-trigger board, and a Raspberry Pi 5.

and reconstruct position and orientation photogrammetrically (Chapter 5). They measure launch conditions — including 3D spin from dimple-pattern rotation — essentially perfectly for simulator purposes, then *model* the flight. Their weakness is the mirror image of radar's: everything downrange of the capture volume is simulated, and club data requires either fiducial stickers on the face or an overhead viewing geometry.

The market's convergent evolution is the single most instructive fact for a new design: *every* high-end vendor has concluded that neither modality suffices alone. TrackMan added cameras to its radar (OERT, Section 4.6); FlightScope added Fusion Tracking cameras; SkyTrak added radar to its camera; Rapsodo pairs radar with impact cameras; Full Swing feeds a camera into its radar ML pipeline. Chapter 6 tabulates the landscape.

1.3 Reading guide

Chapter 2 fixes terminology and coordinate conventions (TrackMan's definitions, the de facto industry standard). Chapter 3 develops the impact physics connecting club delivery to ball launch — the D-plane model, spin generation, and gear effect — which every monitor uses either forward (simulation) or inverse (parameter estimation). Chapters 4 and 5 treat the two sensing families in depth. Chapter 6 surveys the commercial devices. Chapter 7 maps the patent landscape with expiry status. Chapter 8 covers ball-flight aerodynamics and trajectory estimation. Chapter 9 reviews the independent validation literature. Chapter 10 distills the implications for OpenFlight.

Key point

Throughout, we use a right-handed coordinate system for a right-handed golfer: x down the target line, y up, z to the golfer's right. Angles are positive right/up. All club-delivery values are referenced to the moment of *maximum ball compression*; all ball-launch values to the moment of *separation from the face*.

Chapter 2

Parameter Definitions and Conventions

Launch monitors from different vendors disagree partly because they measure different things and partly because they *define* things differently. TrackMan's definitions [65, 60] are the industry reference and are adopted here. The critical subtlety is *when* each quantity is defined: club-delivery parameters at the time of **maximum compression**, ball parameters **immediately after separation**.

2.1 Club-delivery parameters

Table 2.1: Club-delivery parameters (TrackMan conventions). "GC" = the geometric center of the club head.

Parameter	**Definition**	**Notes**
Club speed	Linear speed of the GC just prior to first contact	Not the impact-point speed; the toe moves up to ∼7 mph faster than the heel
Attack angle	Vertical direction of GC motion at maximum compression	+ = hitting up. PGA Tour driver avg $\approx -1.3°$; LPGA $\approx +3°$
Club path	Horizontal direction of GC motion at maximum compression	+ = in-to-out (right of target for RH)
Face angle	Horizontal direction the face normal points, at the center-point of ball contact, at maximum compression	+ = open
Face to path	Face angle − club path	Sign controls curvature
Dynamic loft	Vertical angle of the face normal at the contact point at maximum compression	Differs from static loft via shaft lean/bend, face roll, impact height
Spin loft	3D angle between the club-motion direction (path, attack angle) and the face-normal direction (face angle, dynamic loft)	≈ dynamic loft − attack angle only when face-to-path ≈ 0
Swing plane	Vertical angle of the plane traced by GC motion vs. the horizon	
Swing direction	Horizontal angle of that plane's base vs. the target line	
Low point	Distance from GC at maximum compression to the lowest point of the swing arc	+ = low point ahead of the ball (ball-first contact)
Impact height / offset	Vertical / horizontal strike location relative to face center	Drives gear effect (Section 3.4)
Dynamic lie	Shaft angle vs. horizontal at impact	
Closure rate	Angular rate at which the face is closing in 3D (°/s)	Foresight GCQuad-class only

Two definitional choices deserve emphasis because they are common sources of inter-device disagreement:

1. **Which point on the club is tracked.** TrackMan defines club speed at the *geometric center* of the head and reconstructs that point from the radar "3D silhouette" of the head [67]. A naive Doppler processor instead locks to the strongest or fastest return — often the toe of a driver — and reports a speed several mph high. This single choice explains much of the chronic club-speed disagreement between brands.

2. **Where on the face orientation is evaluated.** Face angle and dynamic loft are defined at the *contact point*, not the face center. On a curved driver face (bulge and roll, Section 3.4) the local normal at a toe strike differs from the center normal by several degrees, so systems that measure face pose but not impact location are systematically biased on off-center hits.

2.1.1 Club path is reference-point dependent too — by about 3° on a driver

The reference-point issue is usually discussed only for club speed, but it applies with equal force to *direction*, and the magnitude is large enough to matter.

For any two points on the rigid clubhead,

$$\boldsymbol{v}_P = \boldsymbol{v}_O + \boldsymbol{\omega} \times \boldsymbol{r}, \qquad \boldsymbol{r} = \boldsymbol{p}_P - \boldsymbol{p}_O, \tag{2.1}$$

so there is a *velocity field* across the head rather than a single path. A reported path is that field sampled at a chosen point. The two candidate points are far apart on a driver: the center of gravity sits roughly 25–50 mm behind the face, and the geometric center that radar tracks is typically within 6 mm of the CG [67].

The industry has split along the sensing modality. Radar systems report path at the geometric center, because that is the point the silhouette reconstruction can locate from behind. Optical systems with face fiducials report it at the *face center*, because that is where the markers are. TrackMan quantifies the resulting gap directly: for a driver, the CG path and the face-center path differ by **approximately 3°**, with the face-center path being the more out-to-in of the two, and the discrepancy shrinks for shorter clubs as the CG-to-face distance falls [60].

Two rotations drive it, both pushing the same way horizontally: the swing arc curves leftward through impact, so a point displaced forward along the arc has its velocity rotated further around it; and face closure about the shaft axis swings a point ahead of the rotation center leftward. The same geometry tilts the face-center velocity slightly *upward* relative to the CG — a shallower attack angle — because the face center sits ahead of the CG on an arc that is turning upward through the bottom.

Key point

Because the offset is $\boldsymbol{\omega} \times \boldsymbol{r}$, it is not a fixed constant: it scales with closure rate and arc tightness. Two players with identical reported path but different release rates do not have the same face-center path. And note that TrackMan's reported face-to-path is already a *hybrid* — face orientation evaluated at the contact point minus velocity direction evaluated at the geometric center — so it is not a physically clean angle under either convention.

Implication for OpenFlight

This is the sharpest available argument for the twist formulation of Chapter E. Estimating $\xi = (\boldsymbol{\omega}, \boldsymbol{v}_O)$ makes path at any point a projection of one fitted object, so both conventions — and the rotation rate that separates them — fall out of the same estimate. OpenFlight should report path at a *declared* reference point, state which one, and expose $\|\boldsymbol{\omega}\|$ alongside it so the user can see how much the choice is worth on that swing. Reporting a bare path number without its reference point is reporting a quantity that is $\sim 3°$ ambiguous on a driver.

2.2 Ball-launch parameters

Table 2.2: Ball-launch parameters (defined immediately after separation).

Parameter	**Definition**
Ball speed	Speed of the ball's center of gravity at separation
Smash factor	Ball speed ÷ club speed
Launch angle	Vertical takeoff angle vs. the horizon
Launch direction	Horizontal takeoff angle vs. the target line
Spin rate	Rotation rate about the (single) spin axis, in rpm
Spin axis	Tilt of the rotation axis relative to the horizon; $-$ = tilted left ⇒ draw for a right-hander
Carry / side / total	Trajectory descriptors; carry is measured to the point at launch elevation
Apex, landing angle	Peak height; descent angle at landing

A golf ball in flight has exactly one rotation vector $\boldsymbol{\omega}$. "Backspin" and "sidespin" are components of that vector, not separate spins. Systems that report sidespin (SkyTrak-style) and systems that report spin axis (TrackMan-style) are related by

$$S_{\text{side}} = S \sin\theta_{\text{axis}}, \qquad S_{\text{back}} = S \cos\theta_{\text{axis}}, \qquad \theta_{\text{axis}} = \operatorname{atan2}(S_{\text{side}}, S_{\text{back}})\,. \tag{2.2}$$

A useful rule of thumb from TrackMan's data: 1° of spin-axis tilt produces roughly 0.7% of carry as side-curve (about 0.7 yd per 100 yd) [44].

2.3 The directness hierarchy

For any launch monitor, each reported parameter falls somewhere on a hierarchy from directly measured to purely modeled. Anticipating the detailed treatment in Chapters 4 and 5, the hierarchy for the two sensing families is summarized in Table 2.3. This table is the skeleton of this whole review.

Table 2.3: Measured vs. derived, by architecture. "M" = measured directly, "D" = derived through a physical model, "E" = estimated/model-fit, "–" = not available.

Parameter	**Radar (outdoor)**	**Radar (indoor)**	**Camera (photometric)**	**Hybrid**
Ball speed	M	M	M	M
Launch angles	M	M	M	M
Spin rate	M[a]	M/E[a]	M[b]	M
Spin axis	D[c]	E	M[b]	M
Carry	M	E[d]	E[d]	E[d]
Club speed	M[e]	M[e]	M[f]	M
Club path / attack	M	M	M[f]	M
Face angle	D[g]	D[g]	M[f]	M
Dynamic loft	D[g]	D[g]	M[f]	M
Impact location	–	–	M[f]	M[h]

[a] Doppler harmonic sidebands; requires surface asymmetry and sufficient flight (Section 4.4.2). [b] Dimple-pattern registration (Section 5.2.2). [c] Inverted from trajectory curvature via the Magnus constraint (Section 4.4.3). [d] Trajectory model integration from measured launch (Chapter 8). [e] Reference-point dependent; silhouette reconstruction on TrackMan. [f] Requires face fiducials (Foresight/Garmin R50) or overhead geometry (Uneekor/ProTee). [g] D-plane inversion from ball launch + path on radar-only units; optically assisted on OERT-class hardware (Section 4.5.2). [h] Markerless via camera on TrackMan 4/iO.

Chapter 3

Impact Physics: From Club Delivery to Ball Launch

Every launch monitor embeds a model of the club–ball collision. Camera systems use it *forward* (sanity-checking and filling gaps); radar systems use it *inverse* (recovering face angle and dynamic loft from measured ball launch). This chapter develops that model.

3.1 The D-plane

Jorgensen [28] observed that at impact two unit vectors determine the collision geometry for a center strike:

- $\hat{\boldsymbol{n}}$, the **face normal**, built from face angle φ_f and dynamic loft λ_d;
- $\hat{\boldsymbol{p}}$, the **club-head velocity direction**, built from club path φ_p and attack angle α.

These two vectors span a wedge-shaped plane — the *D-plane* ("descriptive plane"). Its two governing consequences are:

1. **Launch direction lies in the D-plane**, between $\hat{\boldsymbol{n}}$ and $\hat{\boldsymbol{p}}$, much closer to the face normal.
2. **The spin axis is normal to the D-plane**:

$$\hat{\boldsymbol{\omega}} \propto \hat{\boldsymbol{p}} \times \hat{\boldsymbol{n}}. \tag{3.1}$$

   The ball therefore curves *away from the path, around the face* — the "new ball flight laws."

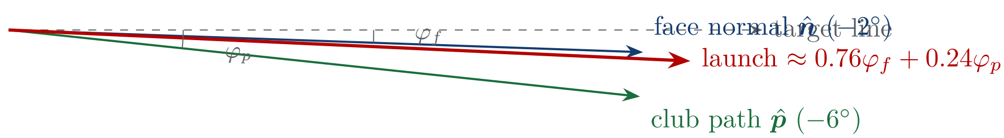

Figure 3.1: Horizontal D-plane geometry (top view, right-handed golfer, out-to-in "fade" delivery). The ball launches close to the face direction; the face-to-path difference tilts the spin axis and curves the ball away from the path.

3.1.1 Face/path weighting of launch direction

Robot and player data (TrackMan's 2009 ball-flight-laws analysis [59]) give the canonical weighting of horizontal launch direction between face angle and club path:

$$\varphi_{\text{launch}} \;\approx\; w_f\,\varphi_f + (1 - w_f)\,\varphi_p, \qquad w_f \approx \begin{cases} 0.76 \pm 0.08 & \text{driver} \\ 0.69 & \text{7-iron} \\ 0.61 & \text{wedge.} \end{cases} \tag{3.2}$$

Caution

The values above are the *horizontal* weights measured by Wood et al. [56] across 157 golfers and 1,575 shots at 720 fps, filtered to strikes within 0.25 in of face centre; a PING Man robot test returned 0.63 for the 7-iron.
The familiar 0.85/0.15 figure comes from TrackMan's *Ten Fundamentals*, which asserts 85% in *both* planes—85% dynamic loft to launch angle, and 85% face angle to initial direction. Against Wood's measurements the two claims fare differently: the vertical claim holds (0.83 ± 0.08), the horizontal one does not (0.76 ± 0.08). This is a live vendor-versus-measurement disagreement, not a rounding difference or a units error.
For OpenFlight this matters directly: inverting Eq. (3.2) to recover face angle amplifies any bias in measured launch direction by $1/w_f$. At $w_f = 0.76$ that is 1.32, not the 1.15 implied by 0.85—roughly twice the error sensitivity.

The weight is loft-dependent: more loft means a more oblique impact, hence a larger tangential momentum component along the path direction. Peer-reviewed impact modeling with normal COR plus tangential compliance reproduces the loft dependence from first principles [57, 56].

The *mechanism* is contested. TrackMan attributes the driver's higher weight to a smoother, lower-friction titanium face; PING rejects this, showing the loft dependence persists at constant μ and that below $\sim 20°$ of incidence *lower* friction moves launch *closer* to the path—the opposite of the friction hypothesis [57]. An implementation that fits w_f empirically should record which explanation its calibration assumes.

Tutelman [70] gives a closed-form version calibrated to TrackMan data. With face-to-path angle A and dynamic loft L, define the total obliqueness Φ (which is precisely the spin loft):

$$\cos\Phi = \cos A \cos L, \qquad f(\Phi) = 0.96 - 0.0071\,\Phi \quad (\Phi \text{ in degrees}), \tag{3.3}$$

and the departure angles relative to the club path are

$$\mathrm{DA}_{\text{vert}} = L\, f(\Phi), \qquad \mathrm{DA}_{\text{horiz}} = A\, f(\Phi), \tag{3.4}$$

with launch angles relative to the target obtained by adding attack angle and path respectively. Note $f(\Phi) \approx 0.85$ at $\Phi \approx 15°$ (driver) and ≈ 0.75 at $\Phi \approx 30°$ (short iron): one friction-calibrated function reproduces both rules of thumb.

3.1.2 Spin-axis tilt

For a center strike the spin-axis tilt follows from the D-plane geometry:

$$\tan\theta_{\text{axis}} \approx \frac{\tan(\text{face-to-path})}{\tan(\text{spin loft})}, \tag{3.5}$$

so 1° of face-to-path tilts the axis $\approx 4°$ with a driver (spin loft ∼12–15°) but only $\approx 2°$ with a mid-iron (spin loft ∼25–30°) [44]. Tuxen's shot-shaping rule for a ball that curves back to the target line: $\theta_{\text{axis}} \approx -2.5\times$ the horizontal launch angle.

Implication for OpenFlight

OpenFlight measures launch direction (K-LD7 horizontal) and club path. Equation (3.2) then yields face angle by inversion: $\varphi_f = (\varphi_{\text{launch}} - (1 - w_f)\varphi_p)/w_f$. Because $1/w_f \approx 1.15$, any systematic launch-direction bias is *amplified* ∼15% in the reported face angle — alignment calibration of the horizontal angle radar is the single most important accuracy investment for club data.

3.2 Ball speed and smash factor

The normal-direction collision is well modeled as a two-body impact with coefficient of restitution e [6, 43]:

$$v_{\text{ball}} = v_{\text{club}} \frac{1+e}{1+m/M} \cos\Phi \,(1 - 0.14\, x_{\text{miss}}), \tag{3.6}$$

with $e \approx 0.83$ at the USGA driver limit (falling with loft and impact speed), ball mass $m = 45.93\,\text{g}$, head mass $M \approx 200\,\text{g}$ (driver), $\cos\Phi$ the obliqueness loss, and the last factor an empirical off-center loss with x_{miss} in inches [69]. The prefactor $(1+e)/(1+m/M) \approx 1.49$ sets the theoretical driver smash ceiling; smash factor falls with loft:

Effective loft	0°	10°	20°	30°
Max smash factor	1.488	1.465	1.398	1.288

Key point

This table is a built-in sanity check: a measured smash factor above the loft-appropriate ceiling indicates a measurement error — most commonly a radar unit reporting toe speed or shaft speed instead of geometric-center club speed. OpenFlight should flag physically impossible smash values instead of displaying them.

3.3 Spin generation

Friction during the oblique compression converts tangential surface speed $v_{\text{club}} \sin\Phi$ into rotation. Real impacts exhibit tangential compliance (the contact patch sticks and stretches like a shear spring) [57, 7], but a practical engineering fit calibrated to TrackMan data [70] is simply

$$S_{\text{total}}\ [\text{rpm}] \approx 160\, v_{\text{club}}[\text{mph}]\, \sin\Phi, \tag{3.7}$$

with the back/side split following from the D-plane direction ratio $d = \tan A/\tan L$: $S_{\text{back}} = S/\sqrt{1+d^2}$, $S_{\text{side}} = d\, S_{\text{back}}$. Cross-checks: ∼200–300 rpm per degree of spin loft for a driver; the "iron loft × 200" rule matches PGA Tour averages. Friction saturates near spin lofts of 45–50° (the "spin-loft cliff"), and wet or grass-contaminated contact reduces spin at fixed spin loft [64].

3.4 Gear effect and impact location

An impact at horizontal offset x from the CG line torques the head about its CG; the recoiling face "gears" the ball the opposite way. The angular-impulse model [68] gives head rotation $\omega_{\text{head}} = x\, m\, v_{\text{ball}}/I_h$ and gear spin

$$s\ [\text{rpm}] = 58{,}830\, \frac{v_{\text{ball}}[\text{mph}]\ C[\text{in}]\ x[\text{in}]}{I_h[\text{g cm}^2]} \approx 16.4\, v_{\text{ball}}[\text{mph}]\ x[\text{in}] \times 100, \tag{3.8}$$

where C is CG depth behind the face (32–47 mm on measured OEM drivers) and $I_h = 4000$–$5800\,\text{g cm}^2$; the ratio I_h/C is nearly constant across modern drivers, which is why the simplified form works to ±2.5%. A toe strike opens the head and imparts *hook* gear spin; a heel strike, slice spin. Vertical gear effect (high-face strikes reduce backspin, low-face strikes increase it) follows the same equation with the vertical CG offset and roll radius.

Driver faces are built curved to exploit this: **bulge** (horizontal radius, typically 12 in) starts a toe miss right so the gear-effect hook curves it back. Worked example [68]: a 1 in toe miss at 150 mph ball speed gets +1354 rpm of bulge-induced slice spin against −2192 rpm of gear hook spin — net 838 rpm hook and ∼10 yd left, versus ∼61 yd offline for a flat face. Sensitivity is extreme: a strike one dimple width (0.14 in) off-center tilts the spin axis $\sim 6°$ on a driver ($\sim 2°$ on a 6-iron) [44]. Irons have shallow CG depth, hence weak gear effect and flat faces.

Implication for OpenFlight

Gear effect is why *impact location* is the most valuable club parameter a launch monitor can add after path/face: without it, a pure D-plane inversion misattributes gear-effect spin-axis tilt to face-to-path. For a radar-only OpenFlight this is a fundamental, quantifiable error source on off-center driver strikes (up to several degrees of apparent face-to-path); Equation (3.8) bounds it, and an optical impact-location add-on (Chapter 10) removes it.

3.5 Vertical plane and the low-point geometry

The identical geometry applies vertically: launch angle sits 75–85% of the way from attack angle toward dynamic loft, and spin loft sets spin magnitude via Equation (3.7). One non-obvious coupling: for a descending strike on an inclined swing plane, the instantaneous path points in-to-out even when the swing direction is square — e.g. hitting 5° down on a 60° plane yields $\approx 2.9°$ of in-to-out path. Radar systems that measure swing plane and attack angle use this relation (path $= f$(swing direction, swing plane, attack angle)) as a consistency constraint among the measured club parameters.

Chapter 4

Doppler Radar Systems

4.1 Continuous-wave Doppler fundamentals

A CW radar transmits a carrier at frequency f_c (wavelength λ) and receives echoes shifted by the Doppler effect. For a target with radial velocity v_r,

$$f_d = \frac{2v_r}{\lambda} = \frac{2f_c v_r}{c}. \tag{4.1}$$

At the 24.125 GHz K-band center frequency used by the OPS243-A (and the Garmin R10, Full Swing KIT, original Mevo), each 1 mph of radial speed produces ≈ 71.7 Hz of shift; at X-band ($\sim$10.5 GHz, TrackMan's long-range subsystem and the Mevo+) the constant is ≈ 31.3 Hz/mph. A spinning, translating golf ball is not a point target: every surface patch has its own radial velocity, so the return is a velocity *spectrum* whose structure carries the spin information (Section 4.4.2).

Band choice is a range-vs-resolution trade. X-band supports long-range full-flight tracking (TrackMan tracks the entire $\sim$6 s flight); 24 GHz gives finer velocity resolution per unit observation time and compact low-power hardware, but consumer 24 GHz units track only $\sim$30 yd of flight [9, 19, 36].

4.2 From velocity sensor to 3D tracker: phase interferometry

A single-channel Doppler radar measures radial speed only. Every serious launch monitor adds *multiple receive antennas* and measures the *phase difference* of the return across receiver pairs (phase-comparison monopulse / interferometry). A wavefront arriving from direction $\hat{\boldsymbol{u}}$ reaches two antennas separated by baseline $\boldsymbol{d}$ with time delay $\tau = (\boldsymbol{d} \cdot \hat{\boldsymbol{u}})/c$, observed after mixing as a phase difference

$$\Delta\varphi = 2\pi f_c \tau \pmod{2\pi} \quad \Longrightarrow \quad u = \frac{\lambda\,\Delta\varphi}{2\pi d}, \tag{4.2}$$

where u is the direction cosine along the baseline. Two orthogonal baselines give azimuth and elevation; the 2π ambiguities are resolved with more than three antennas at staggered spacings (TrackMan US 9,958,527 [53]). Combined with range (from multi-frequency CW phase differences or an FMCW chirp) and range rate (Doppler), the radar produces a full 3D track $(r, \mathrm{az}, \mathrm{el}, \dot{r})$ per epoch. Tuxen's spin-axis patent describes exactly this "phase–phase monopulse comparison ... or interferometry" across three receivers [72]; the Garmin R10 uses a three-receiver 24 GHz array the same way [36]; FlightScope describes its aperture as a phased antenna array [55].

Implication for OpenFlight

OpenFlight's architecture — OPS243-A for speed plus two K-LD7 modules for vertical and horizontal angles — is a physically separated version of what commercial units do on one board: the K-LD7's RADC mode exposes raw ADC data from its two receive patches, and per-bin phase comparison Equation (4.2) yields the angle of each detection. The commercial lesson is that angle accuracy is limited by baseline length, SNR, and mutual calibration; a rigid mechanical mount and a one-time angular calibration against surveyed targets matter more than any software refinement.

4.3 CW + FMCW hybrids

Full Swing's patent US 11,311,789 (published as US 2020/0147470) [8] is the clearest public description of a hybrid launch-monitor radar: two ∼24 GHz transmitters, time-division multiplexed, one CW and one linear-FMCW. The CW channel provides club speed, ball speed, spin, and launch-window angles from Doppler; the FMCW chirp provides direct range to 250 m+, carry, and side displacement from beat-frequency and phase measurements. The antenna layout is deliberately *non-uniform* — four receive antennas symmetric about the axis, two transmit antennas placed asymmetrically — to enrich the phase/frequency structure of the receive signal. FlightScope's X3 similarly advertises "multi-frequency radar with direct distance measurement" [55].

4.4 Ball measurement

4.4.1 Speed and launch angles

Ball speed is read from the dominant Doppler line immediately after impact; launch angles come from the interferometric track over the first meters of flight. These are the most directly measured quantities in the whole parameter set and show the best inter-device agreement in every independent study (Chapter 9). Because the radar sits behind (or above) the ball, the measured radial speed underestimates the true speed by the cosine of the aspect angle; commercial units correct this using the tracked geometry, and the correction is largest for high launch angles and close radar placement.

4.4.2 Spin rate: the harmonic-sideband method

The foundational method is Tuxen's patent US 8,845,442 [71] (EP 1 698 380). A golf ball's dimples, paint, logo, seam, and core asymmetry make its radar cross-section vary periodically as it rotates. A surface feature at radius r contributes a Doppler term

$$f_{d,\text{feature}}(t) = \frac{2}{\lambda}(v_r - r\,\omega \sin \omega t), \tag{4.3}$$

i.e. frequency modulation of the return at the spin frequency $\omega/2\pi$ with deviation $(2/\lambda)r\omega$. The received spectrum therefore shows the main translational Doppler line flanked by **equally spaced harmonic sidebands** at

$$f = f_d \pm n\, f_{\text{spin}}, \quad n = 1, 2, 3, \ldots \qquad \Longrightarrow \qquad S\,[\text{rpm}] = 60\,\Delta f_{\text{sideband}}. \tag{4.4}$$

The patented processing chain: (1) track the central velocity trace through the short-time Fourier transform; (2) detect symmetric sideband peaks; (3) track them over time as spectral traces; (4) *qualify* harmonics by verifying constant equal spacing across consecutive FFT frames; (5) solve the harmonic-number assignment; (6) divide trace offset by harmonic number [71, 62]. In practice a cepstrum or harmonic-product-spectrum estimator finds the comb spacing robustly.

Two practical corollaries: sideband SNR depends on ball-surface asymmetry (pristine two-piece range balls read weakly — the origin of "marked-ball" modes and Titleist RCT balls with embedded metal tags); and the comb yields spin *rate* only — the axis must come from elsewhere.

4.4.3 Spin axis: trajectory inversion

The same patent discloses the standard radar route to spin axis. From the differentiated 3D track, total acceleration $\boldsymbol{A}$ decomposes as $\boldsymbol{A} = \boldsymbol{g} + \boldsymbol{D} + \boldsymbol{L}$ with drag $\boldsymbol{D}$ antiparallel to the airspeed vector and Magnus lift $\boldsymbol{L}$ perpendicular to both the spin axis and the velocity. Projecting out gravity and drag,

$$\boldsymbol{D} = \Big[\frac{(\boldsymbol{A} - \boldsymbol{g}) \cdot \boldsymbol{v}_a}{\|\boldsymbol{v}_a\|^2}\Big]\boldsymbol{v}_a, \qquad \boldsymbol{L} = \boldsymbol{A} - \boldsymbol{g} - \boldsymbol{D}, \tag{4.5}$$

and the orthogonality constraint $\boldsymbol{L} \cdot \hat{\boldsymbol{\omega}} = 0$, stacked over many trajectory points, gives an overdetermined linear system for the spin-axis direction $\hat{\boldsymbol{\omega}}$ [71]. This works well outdoors with seconds of observed flight; indoors, with 2–6 m of flight before the screen, the curvature signal is tiny and the axis becomes an estimate. FlightScope patented a direct alternative — time delays between vertically and horizontally separated receiver pairs yield the axis angle $\Phi = \arctan[S_H T_H/(S_V T_V)]$ [26] — precisely to recover the axis at short range.

4.4.4 Flight: tracked outdoors, modeled indoors

Outdoors, TrackMan-class units report *measured* carry, apex, and landing angle from the full track. Indoors every radar unit observes only the pre-screen segment (TrackMan 4 requires ≥ 4.7 m of throw; FlightScope claims 8 ft suffices with Fusion Tracking) and extrapolates with an aerodynamic model anchored to the vendor's measured-flight database [61, 23]. The Garmin R10 illustrates the consumer floor: it requires $\geq$20 m of visible flight and ball speed above $\sim$90 mph to measure spin at all; otherwise spin is estimated by a fitted model from the measured inputs [35].

4.5 Club measurement

4.5.1 What the radar tracks on the head

The clubhead is a large, complex reflector whose parts move at different speeds (toe faster than heel by up to $\sim$7 mph on a driver). TrackMan defines club speed at the head's *geometric center* and reconstructs that point by "directly measuring the 3D silhouette of the club head" [67]; naive processors lock onto the strongest or fastest return and systematically over-report. Attack angle and club path are the vertical and horizontal direction of the geometric center's velocity at impact; swing plane, swing direction, and low point are properties of the center's trajectory arc through the hitting zone.

4.5.2 Face angle and dynamic loft

Radar cannot see face orientation. On radar-only systems, face angle and dynamic loft are obtained by *inverting the D-plane collision model* (Section 3.1.1): given measured launch direction, club path, and the loft-dependent weighting w_f,

$$\varphi_f = \frac{\varphi_{\text{launch}} - (1 - w_f)\,\varphi_p}{w_f}, \tag{4.6}$$

and analogously dynamic loft from launch angle and attack angle. TrackMan has confirmed these are "derived numbers from direct measurements and a collision model," validated by robot testing [24]. On TrackMan 4/iO with OERT (Section 4.6), the company's position is that club delivery including face angle at the contact point is measured via synchronized radar+camera silhouette tracking rather than back-computed — though the exact split remains proprietary. Consumer units are explicit about the hierarchy: Garmin specifies launch direction at $\pm 1°$ (measured) but face angle at $\pm 2°$ and classifies it as calculated [35].

Key point

The reported "face angle" on a radar launch monitor is, to first order, a rescaled launch direction. It inherits launch-direction bias amplified by $1/w_f \approx 1.15$, plus a gear-effect error on off-center strikes that the model cannot see (Section 3.4). This is not a defect of any particular product but a structural property of the modality.

4.5.3 Known limitations

Independent benchmarking consistently finds club parameters far less accurate than ball parameters for radar (and for camera systems without face markers): the Leach et al. criterion study is the key reference [31]. Irons are harder than drivers (shaft and hosel returns, turf strike clutter, lower speeds); TrackMan's advertised ">90% club-data pickup rate" with OERT is itself an acknowledgment that pickup is not universal [66]. Alignment error translates directly into correlated path/face bias on all units.

4.6 Optically enhanced radar: OERT and Fusion Tracking

TrackMan 4 fuses its dual radar (X-band long-range ball subsystem + 24 GHz high-resolution club/impact subsystem, receivers sampled at 40 ksps) with a built-in camera synchronized in time and space: the camera contributes exact pre-impact ball position, markerless *impact location* on the face, and "4D silhouette" clubhead tracking between radar epochs [66, 9]. The indoor-only TrackMan iO packs a 24 GHz radar with dual cameras — one at up to 4,600 fps for club and ball, one for alignment — under 810–850 nm IR illumination [61]. FlightScope's patented equivalent is Fusion Tracking (US 10,338,209 [25]): the radar's predicted track steers image search windows, the camera's precise angular fixes correct the radar's, and inter-sensor offsets are removed by error minimization. Rapsodo's MLM2PRO pairs radar with 240 fps cameras that read the printed dot pattern on RPT/RCT balls for measured spin rate and axis [37]; Full Swing's KIT feeds a 4K camera into an ML pipeline alongside its CW+FMCW radar [14].

Key point

Every major radar vendor has independently concluded that pure Doppler is insufficient at short range — for spin axis, impact location, and club-face data — and has added optical sensing or instrumented balls. This is the strongest architectural signal in the market for OpenFlight's roadmap.

Chapter 5

Photometric (Camera) Systems

5.1 Imaging architectures

Photometric launch monitors capture a short burst of high-speed, IR-strobed stereo images over a small capture volume beside (or above) the tee — roughly the first 30 cm of ball flight — and reconstruct ball and club kinematics photogrammetrically. The commercial family tree:

- **Foresight Sports** defined the reference floor-unit design. The GC2 (2010) used two cameras ("stereoscopic"), with club data via the add-on HMT module (two more cameras). The GC3 / Bushnell Launch Pro (a Foresight-built rebadge) use three cameras; the GCQuad and QuadMAX use four cameras at the corners of the sensor window, capturing on the order of 200 images of ball and club per shot at burst rates quoted up to 10,000 fps; the ceiling-mounted GCHawk points the same quadrascopic package downward [11, 46]. Onboard IR object tracking detects the teed ball and the inbound clubhead and triggers the burst; intrinsics/extrinsics are factory-calibrated ("no calibration needed") [11].
- **Uneekor** mounts overhead: EYE XO/XO2 look straight down from ∼3 m with two (XO2: three) high-speed IR cameras at 3,000+ fps [73]. The overhead geometry sees the clubhead and face region directly, enabling sticker-free club data and impact-location video replay (Club Optix). The portable EYE MINI reverts to a side view and therefore requires club stickers.
- **ProTee VX**: overhead, two synchronized high-speed cameras with IR illumination and an ML shot-analysis pipeline; sticker-free club and ball data including vertical/horizontal impact point [47].
- **Garmin Approach R50**: three high-speed cameras arranged horizontally in a floor unit; purely optical ball and club measurement, with clubhead data requiring reflective fiducials [15].
- **SkyTrak / SkyTrak+**: the original SkyTrak was photometric and ball-only; the SkyTrak+ added a dual Doppler radar module specifically for club data while the cameras handle ball launch and spin [51].

5.2 Ball measurement

5.2.1 Photogrammetric position and velocity

With factory-known intrinsics K_i and extrinsics $[R_i|\boldsymbol{t}_i]$, each synchronized image set yields a 3D ball-center fix by triangulation: back-project the matched detections as rays and solve the least-squares intersection (or the DLT system $\tilde{\boldsymbol{x}} \times (P\boldsymbol{X}) = 0$), refining by reprojection error. Ball speed is the displacement between timestamped fixes; launch angle and direction are the components of the initial velocity vector in the calibrated target frame. Two error sources dominate: depth error grows quadratically with distance,

$$\sigma_Z = \frac{Z^2\,\sigma_{px}}{f\,B}, \tag{5.1}$$

for baseline B and focal length f — which is why photometric units keep the capture volume small and also exploit the known ball diameter (21.34 mm radius) as an auxiliary range cue; and center-finding bias — fitting the projected *conic* outline rather than taking the blob centroid avoids a systematic offset of up to RZ/f pixels.

5.2.2 Spin: dimple-pattern registration

The flagship photometric capability is markerless 3D spin. Successive frames of the ball are registered on the sphere: the estimator searches the rotation $R \in SO(3)$ that best maps the back-projected surface texture (the dimple pattern, plus any logo or blemish) of frame k onto frame $k+1$ — Foresight brands this *spherical correlation*; Uneekor, *Dimple Optix* [10, 75]. The recovered R gives everything at once:

$$\hat{\boldsymbol{\omega}} = \mathrm{eig}_1(R), \qquad \theta = \arccos\Big(\frac{\mathrm{tr}\,R - 1}{2}\Big), \qquad \omega = \theta/\Delta t, \tag{5.2}$$

i.e. the spin axis is the eigenvector of R with unit eigenvalue and the spin rate follows from the per-frame rotation angle. Reported accuracy is 1–3% of true spin, independent of ball speed. The known failure mode is rotational aliasing at high spin and low frame rate (the $n \cdot 2\pi$ ambiguity), resolved by multi-hypothesis tracking or a trajectory-consistency prior. Where resolution cannot support dimple registration, systems fall back to *marked balls*: Uneekor QED and Rapsodo (RPT balls) track printed high-contrast features, giving closed-form rotation from ≥ 3 correspondences via the Kabsch/Horn algorithm.

5.3 Club measurement

5.3.1 Fiducial markers: the Foresight approach

Foresight floor units require retroreflective **fiducial dots** on the clubface. The dot count maps directly to the data tier [12, 13]: one dot suffices for club speed, path, and attack angle (tracking a single 3D point through the capture volume); *four dots* — placed on the vertical centerline, equidistant from the horizontal centerline — define the face plane and enable face angle, dynamic loft, lie, closure rate (°/s), and **impact location**. Mechanically: the IR strobe makes the dots bloom in every frame; triangulating the dot constellation across cameras and frames yields the 6-DOF pose of the face through impact. Club speed is the pose translation rate; path and attack angle are the velocity direction; face angle and dynamic loft are the face-plane orientation at contact; impact location is the measured ball position expressed in the face frame. The GC3/Launch Pro tier delivers speed, path, attack, face angle, and impact via the same stickers but omits dynamic loft, lie, and closure rate — these remain GCQuad/QuadMAX exclusives [46].

5.3.2 Sticker-free overhead tracking

Uneekor's overhead cameras segment the clubhead silhouette directly — no markers — because the top-down view presents the head against the mat with controlled IR illumination; the systems also record actual high-speed impact video (face-on via Club Optix), from which impact location is read [73]. ProTee VX claims the same sticker-free club set (speed, path, face, attack, dynamic loft, lie, vertical and horizontal impact point) from its overhead ML pipeline [47]. The trade-off is installation: overhead units need ∼3 m ceilings, rigid mounting, and a floor-chart calibration.

5.3.3 Calibration

Factory-rigid multi-camera rigs ship calibrated; the user aligns only the target line. Overhead systems require field calibration: Uneekor's procedure places a printed calibration chart on the mat, levels it, squares it to the screen, and aligns its crosshair to overlays on the live camera feeds — establishing the ground plane, hitting zone, and target-line extrinsics [4]. This is the pattern a DIY overhead build should copy.

5.4 The indoor argument, and what cameras cannot do

Photometric systems measure actual launch conditions — including 3D spin — in the first foot of flight; nothing is lost when the ball hits a screen 2.5 m away. Radar indoors must infer from a truncated flight. This is the structural reason camera units dominate indoor simulation while radar dominates outdoor practice: each modality measures what the other models [21]. The camera's blind spots are downrange (all flight is simulated from launch) and, for floor units, the club face itself without stickers — the same face-visibility problem radar has, solved with markers instead of models.

5.5 DIY and open-source photometric systems

PiTrac [45] is the flagship open-source photometric build and the most relevant external project to OpenFlight. Instead of multi-kfps cameras it uses *strobed multi-exposure capture*: an IR LED array pulses several times within one long exposure of a ∼$50 Raspberry Pi Global Shutter camera, freezing multiple ball images in a single frame. One camera watches the teed ball and detects launch; the second captures the strobed flight images; a custom PCB sequences trigger and strobe. Processing (C++/OpenCV): Hough-circle detection locates ball images; strobe-timestamped positions give speed and launch angles; 3D spin on all three axes is solved by searching candidate rotations that best register the dimple imagery between exposures — Gabor-filter dimple enhancement is documented in the adjacent patent literature [58]. Total BOM ≈$250–300. The architecture is a direct descendant of the now-expired Wintriss patents (Section 7.2) and of the GC2's flash-multi-exposure technique.

Implication for OpenFlight

PiTrac proves that commodity global-shutter sensors + IR strobing deliver photometric-grade launch and spin measurement at OpenFlight's price point. For OpenFlight the natural division of labor is: keep the radar core for speed/trigger robustness and outdoor use, and adopt a PiTrac-style strobed camera as the spin/impact-location module indoors — the same fusion direction every commercial vendor has taken, from the opposite starting modality.

Chapter 6

Commercial System Survey

Table 6.1 summarizes the sensing architecture of the major systems; the notes that follow give the technically salient details per device, with sources in the bibliography.

Table 6.1: Launch monitor architecture survey (2026). "Club source" distinguishes directly measured face data (optical) from D-plane-inverted or model-estimated face data.

System	Primary sensing	Spin method	Club data source	Impact location
TrackMan 4	Dual radar (X-band + 24 GHz) + camera (OERT)	Doppler sidebands	Radar silhouette + camera	Yes (markerless, OERT)
TrackMan iO	24 GHz radar + 4,600 fps camera	Camera + radar	Radar + camera	Yes (markerless)
FlightScope X3	Phased-array multi-freq. radar + Fusion cameras	Doppler (dielectric-lens)	Radar; face derived	No
FlightScope Mevo+	X-band 10.5 GHz phased array (+ Pro cameras)	Doppler; stickers indoors	Radar; face derived	No
Garmin R10	24 GHz, 3-receiver CW	Doppler ($\geq$20 m flight) or model	Radar; face calculated	No
Garmin R50	3 high-speed cameras	Dimple imaging	Optical (fiducials)	Yes
Full Swing KIT	24 GHz CW+FMCW + 4K camera ML	Doppler + ML	Radar + ML	No
Rapsodo MLM2PRO	Radar + 2×240 fps cameras	Marked ball (RPT/RCT)	Radar; face derived	Video only
Foresight GCQuad / QuadMAX	4 cameras, IR strobe	Spherical correlation	Optical (4 fiducials)	Yes (measured)
GC3 / Launch Pro	3 cameras, IR strobe	Spherical correlation	Optical (fiducials; add-on)	Yes
GCHawk	4 cameras, ceiling	Spherical correlation	Optical	Yes
Uneekor EYE XO2	3 overhead IR cameras, 3,000+ fps	Dimple Optix	Optical, sticker-free	Yes + video
Uneekor EYE MINI	2 cameras, portable	Dimple Optix	Optical (stickers)	Yes
ProTee VX	2 overhead cameras + ML	Dimple imaging	Optical, sticker-free	Yes
SkyTrak+ / ST MAX	Cameras (ball) + dual radar (club)	Photometric	Radar; face derived	No
OpenFlight (current)	OPS243-A 24 GHz + 2×K-LD7 + sound trigger	Doppler I/Q buffer (~50–60% detect)	Radar; no face data yet	No
PiTrac (DIY)	2 Pi GS cameras + IR strobe	Dimple registration	None yet	No

6.1 Radar-first systems

TrackMan 4 / iO. The reference radar architecture: two synchronized radar subsystems (long-range X-band at the corners for full ball flight; higher-frequency 24 GHz at the center for club and impact), receivers sampled at 40 ksps to pin the impact instant, fused with an OERT camera [9, 66, 61]. Reports 27+ parameters including swing plane/direction, low point, dynamic lie, and D-plane-derived face data; requires ≥ 4.7 m of indoor throw. The iO is the indoor-optimized repackaging: 24 GHz radar, 4,600 fps club/ball camera, IR illumination, no minimum-space requirement.

FlightScope X3 / Mevo+. Phased-array Doppler (X3: multi-frequency with direct ranging; Mevo+: X-band 10.5–10.55 GHz per FCC filings [19]) with Fusion Tracking cameras on the X3 and Mevo+ Pro. Spin measured via the patented dielectric-lens phase-demodulation route (Section 7.3); metallic stickers recommended for short indoor flights. FlightScope's 2022 win against TrackMan at the German Federal Court of Justice, after losing the 2013 EP 1 698 380 case, bookends two decades of radar-spin litigation between the two [18].

Garmin Approach R10. The instructive budget case: a 24 GHz three-receiver CW radar that directly measures only the safe primitives — ball speed, launch angles, club speed, club path — and models the rest. Spin requires $\geq$20 m of flight and >90 mph ball speed (or an RCT marked ball indoors); face angle is explicitly a calculated value at ±2° [35, 36]. Garmin's step-up R50 abandons radar entirely for three cameras — a telling modality switch for indoor accuracy.

Full Swing KIT. The patented CW+FMCW dual-mode 24 GHz radar (Section 4.3) with an ML vision assist from its 4K camera; 16 parameters; third-party testing places it within 1–2% of TrackMan/GCQuad in most conditions [14].

Rapsodo MLM2PRO. Radar for speed/launch plus two 240 fps cameras that read printed RPT/RCT ball markers for measured spin rate and axis (claimed within 1% of reference units) — the cheapest route to *measured* spin, at the cost of proprietary balls [37].

6.2 Camera-first systems

Foresight GCQuad / QuadMAX / GCHawk. Quadrascopic IR-strobed capture (∼200 images/shot); spherical-correlation spin; four-dot fiducial club measurement with closure rate and measured impact location (Section 5.3) [11, 12]. Robot testing shows the class-leading spin repeatability (Chapter 9).

GC3 / Bushnell Launch Pro. Identical triscopic hardware in two brands; club data is a paid add-on tier using the same stickers [46].

Uneekor EYE XO2 / EYE MINI / ProTee VX. Overhead sticker-free club measurement (XO2, VX) versus portable sticker-based (EYE MINI); Dimple Optix markerless spin; impact video replay [73, 47].

SkyTrak+ / ST MAX. The camera-to-radar convergence case: the original photometric ball-only SkyTrak gained a dual-radar club module and ML fusion in the SkyTrak+ [51].

6.3 Open-source systems

OpenFlight (radar-first: OPS243-A I/Q rolling buffer + sound trigger + K-LD7 interferometric angle radars) and **PiTrac** (camera-first: strobed multi-exposure global-shutter imaging) occupy the two ends of the same spectrum the commercial market spans, at roughly 1/10th the hardware cost. Neither yet measures club face data; both have clear, patent-informed paths to it (Chapter 10).

Chapter 7

Patent Landscape

The strongest public documentation of proprietary launch-monitor methods is the patents themselves; they are cited throughout this review as technical sources. This chapter organizes them by assignee and closes with a freedom-to-operate (FTO) map. Expiry estimates follow the 20-years-from-priority rule plus patent-term adjustment (PTA), using Google Patents' anticipated-expiration data where shown; *none of this is legal advice, and claim-by-claim analysis by counsel is required before any commercial decision.*

7.1 TrackMan A/S (Interactive Sports Games A/S) — Fredrik Tuxen

TrackMan's public patent list enumerates 45 US patents covering full-trajectory radar tracking, club delivery data, radar spin rate and spin axis, markerless impact location, and OERT fusion [63].

US8845442B2 — "Determination of spin parameters of a sports ball." Priority March 3, 2005; PTA-extended expiry ∼May 2029 [71]. *The* radar-spin patent: harmonic sidebands equally spaced at the spin frequency around the central Doppler line, with the trace-tracking/qualification chain of Section 4.4.2; spin axis from the Magnus-lift orthogonality constraint (Equation (4.5)). The European sibling EP 1 698 380 was upheld by Germany's Federal Court of Justice and grounded TrackMan's 2013 Düsseldorf win against FlightScope's distributor. Continuation US10393870B2 (same disclosure) expires December 2026.

US8085188B2 / US9857459B2 / US10473778B2 — target-deviation family. Priority July 2004. Camera rigidly mounted to the radar; user taps a target in the image; system reports launch-to-target deviation with automatic coordinate transforms. US9857459 lapsed 2022; the siblings expire 2026–27 — a safe UX pattern to adopt shortly.

US10850179B2 (+ US11446546, US11938375) — spin axis. Direct spectral/interferometric spin-axis determination from per-receiver-pair phase differences; also documents the three-receiver monopulse architecture [72].

US10989791B2 / US11828867B2 — radar+imager fused tracking; US11619708B2 / US12517218B2 The actual OERT-supporting families; active into the late 2030s. (Note: the frequently mis-cited "radar + image data 3D tracking" patents US10596416/US11697046/US12128275 belong to **Topgolf Sweden AB (Toptracer)**, not TrackMan — see Chapter D.)

US10315093B2 — trajectory illustration. The broadcast "tracer" overlay (radar track rendered into calibrated video); to 2030.

US11086005B2 — multi-bay tracking. Toptracer-style back-extrapolation of tracks to assign shots to bays; to ∼2036.

7.2 Foresight Sports (WAWGD) and the Wintriss lineage

Foresight Sports is WAWGD, Inc.; the foundational photometric patents it holds (via Wawgd Newco LLC) were invented at Wintriss Engineering by Christopher Kiraly (a Foresight co-founder) — and these same numbers appear on Uneekor's license list following the September 2024 Foresight–Uneekor license agreement [3].

US7292711B2 — "Flight parameter measurement system." Priority June 6, 2002; **expired April 2025** [29]. The blueprint single-camera photometric monitor: factory per-pixel 3D calibration; accelerometer leveling; microphone + small radar horn joint trigger; strobe-lit sequential images; ball center/diameter with known ball size giving 3D position per frame; and *markerless spin* by iterative rotation/correlation of natural surface features (dimples, blemishes) after glint removal and lighting normalization.

US7324663B2 — sibling "smart camera" patent. Same spec, self-triggering from in-FOV motion; **expired August 2025**.

US7497780B2 / US7641565B2 — integrated monitor and ball-placement detection. Priority 2006; expire ∼2027. The GC2-style UX: optical ball-find, LED placement guidance, frame-differencing launch detection, mixed-mode capture, on-device display. Foresight enforced this portable-monitor family as recently as 2024 (Uneekor settlement).

Later WAWGD filings. Applications on measuring club path and face orientation before/at/after impact — the four-dot GCQuad fiducial system — remain active; treat quad-camera + reflective-dot face measurement as protected.

7.3 FlightScope / EDH — Henri Johnson

US9868044B2 — "Ball spin rate measurement." Priority January 2013; expires ∼2034 [27]. The engineered alternative to TrackMan's sidebands after the 2013 loss: the ball's cover acts as a *dielectric lens* ($n \approx 1.8$) magnifying far-side surface features in the microwave return; phase demodulation (PLL) yields repeating bipolar pulses as features sweep the magnification zone; an FFT extracts the periodicity (seam symmetry doubles the modulation rate, corrected in software).

US10775492B2 — "Golf ball spin axis measurement." Priority December 2013; expires ∼2035 [26]. Direct axis measurement from time delays between perpendicular receiver pairs: $\Phi = \arctan[S_H T_H/(S_V T_V)]$ — works at short indoor flights where trajectory inversion fails.

US10338209B2 — Fusion Tracking. Priority 2015; expires ∼2037 [25]. Radar+camera fusion with checkerboard camera calibration, Doppler-simulator radar alignment, radar-steered image search windows, offset removal by error minimization.

7.4 Acushnet (Titleist) — the stereo foundation

The deepest prior art for camera-based monitors; the early family is entirely expired [17, 16, 30].

US5501463A (1992, expired). Two shuttered cameras at ∼22°, double-strobed ∼800 µs apart; three retroreflective dots on the clubhead, six on the ball; triangulation yields 3D clubhead velocity, attack angle, path, face orientation, and *contact location on the face*.

US6500073B1 (1992 priority, expired). Stereo pair + sound trigger + six ball dots; position and orientation at two instants give velocity and angular velocity; numerical flight integration (drag/Magnus/gravity) gives carry and roll — the classic tour package.

US6758759B2 (2001, expired 2022). Dual two-camera monitors (club pre-impact, ball post-impact); magnetic fixture calibrates the head's geometric center; complete recipe for measured face angle and impact location with marker stickers.

US7143639B2 family (2004 priority; parent expired 2024). Portable four-camera unit with speed-adaptive strobe timing via FPGA lookup, dichroic marker discrimination, optical club/ball fingerprinting; continuations US8500568B2/US8556267B2 (hardware integration claims) run to ∼2030–31.

US10668350B2 (2017; to ∼2038). "True 3D" stereo/light-field capture at 1,000–10,000+ fps with sub-10 µs exposures and per-frame *xyz* measurement.

US6186002B1. USGA-adjacent method for extracting C_D/C_L from measured trajectories — a template for calibrating an open ball-flight model [33].

7.5 Korean ecosystem: Creatz/Uneekor and Golfzon

Uneekor's patent list mixes owned Creatz patents (US10247553, US9752875, US9605960, US9448067, US10587797, US10776929, US11191998, US12008770 — including the Dimple Optix markerless-spin engine) with the licensed Foresight/Wintriss numbers [74]. US10247553B2 (Creatz; to ∼2032) claims a sectioned start-sensor detecting the actual ball starting position for simulation. Golfzon's US9242158B2 (to ∼2032) claims the latency-hiding two-stage pipeline — start the simulated trajectory from fast "first ball information" (∼100 ms), refine in flight when the slower spin estimate arrives (∼200 ms) [79].

7.6 Freedom-to-operate map

Key point

Three strategic conclusions. (1) A *camera-first* open design has a wide-open, recently expired foundation (Wintriss + Acushnet) covering mono/stereo photometric launch measurement, markerless spin, and marker-based measured club-face data. (2) *Radar spin* is the most encumbered corner: the TrackMan sideband family runs to ∼2029 and FlightScope's alternatives to ∼2034–35; pure speed/launch-angle radar (the Garmin R10 recipe) is safe. (3) *Radar+camera fusion* is the most actively patented current frontier — the two giants have litigated each other in both directions — and warrants the most care for any hybrid OpenFlight roadmap. Note also that a non-commercial AGPL project is not immune: US patent infringement does not require sale, and downstream commercial users inherit the exposure.

Table 7.1: FTO summary for an open-source launch monitor (US perspective, mid-2026). Verify claim-by-claim with counsel before commercial use.

Technique	**Controlling patents**	**Status**
CW Doppler speed measurement	—	Ancient art; free
Mono-camera photometric launch + markerless dimple-correlation spin	US7292711 / US7324663	**Expired 2025; free**
Stereo + retroreflective dots: club face, path, impact location	US5501463 / US6500073 / US6758759 / US7143639	**Expired; free**
Target-tap deviation UX	US8085188 family	Expires 2026–27
Radar spin via harmonic sidebands	US8845442 (US10393870)	To ∼2029 (Dec 2026)
Radar spin via dielectric-lens phase demodulation	US9868044	To ∼2034
Direct radar spin axis (perpendicular receiver pairs)	US10775492	To ∼2035
Radar+camera fusion tracking	US10338209; TrackMan OERT family	To ∼2037+
Integrated ball-find/placement/trigger workflow	US7497780 / US7641565	To ∼2027
Video tracer overlay from sensor track	US10315093	To ∼2030
Two-stage sim latency hiding	US9242158	To ∼2032
Start-position sensing	US10247553	To ∼2032
Light-field / true-3D capture claims	US10668350	To ∼2038
Multi-bay back-extrapolation attribution	US11086005	To ∼2036

Chapter 8

Ball Flight Models and Trajectory Estimation

Both sensing families depend on an aerodynamic model: cameras integrate it forward from measured launch to get carry; radars fit it to partial trajectories to extract spin axis and to extrapolate indoor flights.

8.1 Equations of motion

With air density ρ, ball radius $R = 21.34\,\mathrm{mm}$, mass $m = 45.93\,\mathrm{g}$, cross-section $A = \pi R^2$, and airspeed $\boldsymbol{v}_a = \boldsymbol{v} - \boldsymbol{v}_{\mathrm{wind}}$:

$$m\,\frac{d\boldsymbol{v}}{dt} = -\tfrac{1}{2}\rho A C_D \|\boldsymbol{v}_a\|\,\boldsymbol{v}_a \;+\; \tfrac{1}{2}\rho A C_L \|\boldsymbol{v}_a\|^2\,(\hat{\boldsymbol{\omega}} \times \hat{\boldsymbol{v}}_a) \;+\; m\boldsymbol{g}, \tag{8.1}$$

with the drag and lift coefficients functions of Reynolds number $\mathrm{Re} = 2R\|\boldsymbol{v}_a\|/\nu$ and spin ratio $S = R\omega/\|\boldsymbol{v}_a\|$. Fourth-order Runge–Kutta integration is ample; carry is evaluated where the trajectory returns to launch elevation. Tilting the spin axis by θ rotates the Magnus force off vertical, giving lateral acceleration $(F_M/m)\sin\theta$ — the mechanism behind the 0.7%-per-degree side-curve rule and Equation (2.2).

8.2 Aerodynamic coefficient data

Bearman & Harvey (1976) [2]. The canonical wind-tunnel dataset on spinning golf-ball models across the flight envelope. Dimples drop the critical Reynolds number to $\sim 5 \times 10^4$; post-critical $C_D \approx 0.25$ is nearly Re-independent; C_L rises with spin ratio from ≈0.08 to 0.25 over $S \approx 0.02$–0.3; hexagonal dimples outperform round ones.

Smits & Smith (1994) [52]. The parameterized model most widely used in simulators:

$$C_D = C_{D1} + C_{D2}\,S + C_{D3}\sin\!\big(\pi\,\tfrac{\mathrm{Re}-A_1}{A_2}\big), \qquad C_L = C_{L1}S^{0.4}, \tag{8.2}$$

with $C_{D1} = 0.24$, $C_{D2} = 0.18$, $C_{D3} = 0.06$, $A_1 = 9 \times 10^4$, $A_2 = 2 \times 10^5$, $C_{L1} \approx 0.54$, plus a spin-decay law with time constant $\approx 24\,\mathrm{s}$ at $100\,\mathrm{mph}$ (roughly 4%/s early in flight).

Quintavalla / USGA Indoor Test Range (2002) [48]. Six-term polynomial C_D/C_L models in Re and S fitted from trajectory photography on the USGA ITR; captures the low-speed end-of-flight regime better than wind tunnels; the basis of USGA Overall Distance Standard conformance testing. The companion method patent US6186002 [33] — determining coefficients from measured trajectories — is the template for calibrating an open model against real flights.

Implication for OpenFlight

OpenFlight should ship Smits–Smith (Equation (8.2)) with spin decay as the default flight model, structured so a Quintavalla-style six-term fit can drop in, and calibrate against measured outdoor trajectories (or MLM2PRO reference data) using the US6186002 trajectory-fitting approach. Carry disagreements between simulators are dominated by model choice, not launch measurement — version the model and record it with every shot log.

8.3 Trajectory estimation and filtering

The estimation core of a radar launch monitor is a nonlinear filtering problem. State $\boldsymbol{x} = (\boldsymbol{p}, \boldsymbol{v}, S, \theta_{\text{axis}})$; process model = Equation (8.1); measurement models: radar $h(\boldsymbol{x}) = (r, \text{az}, \text{el}, \dot{r})$ with $\dot{r} = \boldsymbol{v} \cdot \hat{\boldsymbol{p}}$, camera $h(\boldsymbol{x})$ = pixel projections. An extended (or unscented) Kalman filter runs the standard recursion; launch parameters are then obtained by *smoothing and back-extrapolation* — fit the entire observed arc and evaluate the state at the moment of face separation, which is far more robust than differencing the first noisy fixes. Including θ_{axis} in the state makes spin-axis-from-curvature (Section 4.4.3) fall out of the filter naturally, and per-measurement confidence weighting is the clean way to fuse heterogeneous sensors (OPS243-A speed, K-LD7 angles, future camera fixes).

Implication for OpenFlight

OpenFlight currently correlates K-LD7 angle bursts with the OPS243-A impact timestamp and reads angles from single detections. Migrating to a short EKF smoother over the full K-LD7 ring-buffer burst — even 10–20 detections over 50 ms — would use all available data, reject outlier bins, and yield launch angles with quantified covariance. The same filter skeleton later absorbs camera measurements unchanged.

8.4 Indoor extrapolation and its errors

Indoors, every system integrates Equation (8.1) from launch conditions. Error propagation is dominated by spin uncertainty: at driver speeds, ±300 rpm maps to roughly ±4–6 yd of carry and ±1 yd of side; a ±2° spin-axis error at 2,500 rpm maps to ±3–4 yd of side at 250 yd carry. This is why the accuracy literature (Chapter 9) finds spin to be the fragile channel indoors for radar, and why camera systems — which measure spin directly — win the indoor comparison structurally.

Chapter 9

Accuracy: The Independent Evidence

9.1 Peer-reviewed validation

Leach, Forrester, Mears & Roberts (2017) [31] remains the key criterion study: 240 shots (driver, 7-iron, wedge) measured simultaneously by a TrackMan Pro IIIe, a Foresight GC2+HMT, and a four-camera 5,400 fps optical reference. Findings: ball parameters agreed well on both devices (TrackMan clubhead-speed median difference −0.4 mph; ball speed +0.2 mph; launch angle 0.0°), but **club parameters were materially worse for both** — the authors endorse ball data for research use and advise caution on club data. This is the empirical face of the directness hierarchy of Section 2.3: what is measured agrees; what is derived diverges.

TrackMan 4 indoor reliability (J. Sports Sciences, 2024) [80]: within- and between-session reliability in high-level golfers indoors — ICC 0.99 for club speed, 0.97–0.99 for ball speed, but **spin-rate ICC as low as 0.02–0.60**: even the reference radar's indoor spin channel is fragile when flight is truncated.

A 2025 Mevo+ vs. TrackMan 4 indoor agreement study exists [1] (abstract-level access only at review time), extending the same pattern down-market.

9.2 Robot and industry testing

Golf Laboratories robot testing (reported via Foresight and third parties) puts GCQuad center-strike spin standard deviation at ≈82 rpm versus ≈175 rpm for TrackMan 4 — the photometric spin advantage in its clearest form [20]. MyGolfSpy's indoor comparison against a GCQuad reference found the camera-assisted MLM2PRO among the tightest budget units for launch, spin, and carry, with radar-only units (R10, original Mevo) trailing indoors [38]. Manufacturer-quoted specs bracket the market: ±1 mph ball speed and ±1° launch angles are common claims; club-face claims (±2°, "calculated") are visibly weaker.

9.3 What drives inter-device disagreement

Synthesizing the validation literature and the architecture analysis:

1. **Reference-point differences** in club speed (geometric center vs. fastest return) — several mph of systematic spread.
2. **Derived face data**: D-plane inversion amplifies launch-direction bias by ∼1.15× and cannot see gear effect, so radar face angle diverges from optical face angle most on off-center strikes.
3. **Indoor spin**: sideband SNR collapses on short flights and clean balls; systems silently switch to estimation, and estimated spin feeds the carry model.

4. **Flight-model differences**: identical launch conditions produce different simulated carries across vendors; this is a modeling disagreement, not a measurement one.

5. **Alignment**: a unit misaligned to the target line biases path, face, and launch direction coherently — the cheapest error to fix and the most common in practice.

Implication for OpenFlight

For OpenFlight validation against the MLM2PRO (the project's reference instrument): compare ball speed and launch angles directly; compare spin only on RPT/RCT marked balls where the MLM2PRO's measurement is camera-based; expect club-speed offsets from reference-point differences and calibrate a per-club correction rather than chasing agreement; and log raw I/Q captures so spin-detection improvements can be replayed against historical shots.

Chapter 10

Implications for OpenFlight

OpenFlight's current architecture — OPS243-A 24 GHz CW Doppler with rolling-buffer I/Q capture, a hardware sound trigger, and two K-LD7 interferometric angle radars, on a Raspberry Pi 5 — is a minimal but honest instance of the commercial radar architecture. This chapter distills the review into design guidance, ordered by leverage.

10.1 Ball data

1. **Ball speed** is already the strongest channel: the Doppler line is unambiguous, and the OPS243-A's ±0.5% class accuracy matches commercial practice. Add the aspect-angle (cosine) correction using the measured launch angles — at 3–5 ft behind the tee with a driver launch of 12° the radial underestimate is small but systematic.
2. **Launch angles**: treat the K-LD7 bursts as a short trajectory, not a single detection — an EKF smoother over the ring buffer (Section 8.3) with back-extrapolation to the impact timestamp is the single highest-leverage software upgrade for angle quality.
3. **Spin rate**: the current ∼50–60% detection rate from I/Q sideband analysis is consistent with the physics: sideband SNR depends on ball-surface asymmetry and observation time (Section 4.4.2). Improvements in order of cost: longer observation windows (later trigger cutoff); cepstrum/harmonic-product comb estimation with the patent-style equal-spacing qualification; logo-ball or marked-ball guidance for indoor use; and honest fallback — report estimated spin (from club type + speed + loft priors, clearly flagged) when no comb is found, as Garmin does. *Patent caution:* the harmonic-sideband method is claimed by US8845442 until ∼2029 (US member); for a hobby project this is a risk-management judgment, but the roadmap should note the family's expiry dates (Section 7.6).
4. **Spin axis** is currently out of reach of the radar: indoor flights are too short for trajectory inversion, and the direct receiver-pair method is FlightScope-patented to ∼2035. The practical route is optical (below) or a flagged model estimate from face-to-path.

10.2 Club data

1. **Club speed**: define and document the reference point. The OPS243-A sees a smear of head/shaft returns; gating the pre-impact spectrum and taking a fixed percentile of the velocity distribution (rather than the maximum) approximates a stable reference and avoids the toe-speed inflation that plagues naive processors (Section 4.5). Validate the smash factor against the loft-appropriate ceiling (Section 3.2) and flag violations instead of displaying them.
2. **Club path** (K-LD7 horizontal) and **attack angle** (vertical, if geometry allows) are legitimate direct measurements — the same primitives the Garmin R10 measures. Alignment calibration dominates their accuracy.

3. **Face angle**: implement the D-plane inversion (Equation (4.6)) with the loft-dependent weight (Equation (3.3)) — this is exactly what radar-only commercial units report — but label it *derived*, propagate its error (1.15× launch-direction bias), and suppress it on detected off-center strikes once impact location is available. The D-plane forward model doubles as the simulator's launch generator, so one well-tested module serves both directions.

4. **Impact location** requires optics. The expired Acushnet stereo art (Section 7.4) and Wintriss mono-camera art (Section 7.2) provide complete public-domain recipes; a PiTrac-style Pi Global Shutter camera with IR strobing is the natural hardware. Even a single overhead camera reading face stickers (Acushnet US6758759 recipe, expired) would convert OpenFlight's face angle from derived to measured.

10.3 Architecture roadmap

1. **Phase 1 (radar-only, current)**: EKF smoothing, comb-based spin with honest fallback, D-plane-derived face data with flags, smash-factor sanity gates, alignment-calibration procedure. Ships the Garmin-R10 feature set with open data.

2. **Phase 2 (optical spin/impact module)**: one or two Pi Global Shutter cameras + IR strobe PCB (PiTrac-proven, ∼$100 incremental). Camera measures launch angles, 3D spin (dimple registration, public-domain per expired US7292711), and impact location; radar keeps trigger, speed, and outdoor robustness. This is the same fusion every commercial vendor converged on — but built from the expired-art side of the patent map. *Care:* avoid the still-active claims around integrated ball-find/LED-guidance UX (to ∼2027) and radar-steered image search windows (FlightScope US10338209).

3. **Phase 3 (fusion)**: a single EKF consuming OPS243-A speed, K-LD7 angles, and camera fixes, with per-sensor covariances; flight model per Chapter 8 with versioned coefficients; validation protocol against the MLM2PRO per Chapter 9.

4. **Phase 4 (measured club delivery)**: IWR6843 with custom chirp firmware feeding the screw-theoretic rigid-body estimator of Chapter E — per-detection Doppler rows solved for the club's twist, smoothed on $SE(3)$, and projected into club speed at a declared reference point, measured path and attack angle, closure rate, and an ISA-defined swing plane. This moves path and attack angle from *derived* to *measured* in Table 2.3; face angle and impact location remain with the optical module.

10.4 Reporting honesty as a feature

The single clearest lesson of the review: commercial marketing blurs the measured/derived boundary, and the validation literature repeatedly punishes the derived quantities. An open-source instrument can win trust by doing the opposite — tagging every reported parameter with its provenance (measured / derived / estimated), its uncertainty, and the model version used. Table 2.3 is effectively the schema for that tagging.

Appendix A

Live Reference Library

This appendix collects every substantive source consulted for this review as a clickable link, organized by category. The four raw research dossiers in `tech-review/research/` pair each of these with the specific claims they support.

A.1 Patents (Google Patents / USPTO)

TrackMan A/S (Fredrik Tuxen)

- Full portfolio index: `https://patents.justia.com/inventor/fredrik-tuxen` and `https://www.trackman.com/legal/patents`
- US8845442 — Determination of spin parameters of a sports ball (harmonic-sideband spin; Magnus spin-axis inversion): `https://patents.google.com/patent/US8845442B2/en`
- US10393870 — continuation of the above: `https://patents.google.com/patent/US10393870B2/en`
- US10850179 — spin axis via receiver-pair interferometry: `https://patents.google.com/patent/US10850179B2/en`
- US8085188 / US9857459 / US10473778 — target-deviation (tap-a-target) family: `https://patents.google.com/patent/US8085188B2/en`, `https://patents.google.com/patent/US9857459B2/en`, `https://patents.google.com/patent/US10473778B2/en`
- US10315093 — trajectory tracer overlay: `https://patents.google.com/patent/US10315093B2/en`
- US11086005 — multi-bay range tracking: `https://patents.google.com/patent/US11086005B2/en`
- TrackMan's own spin-patent explainer: `https://blog.trackmangolf.com/patent-measuring-the-spin-of-a-sport`

FlightScope / EDH (Henri Johnson)

- US9868044 — dielectric-lens spin measurement: `https://patents.google.com/patent/US9868044B2/en`
- US10775492 — direct spin-axis via perpendicular receiver pairs: `https://patents.google.com/patent/US10775492B2/en`
- US10338209 — Fusion Tracking (radar+camera): `https://patents.google.com/patent/US10338209B2/en`
- Company history: `https://athlonsports.com/golf/the-flightscope-story-tracking-the-history-of-flight-`

Foresight Sports / Wintriss Engineering (Kiraly)

- US7292711 — flight parameter measurement system (**expired 2025**; markerless dimple-spin blueprint): `https://patents.google.com/patent/US7292711B2/en`
- US7324663 — smart-camera sibling (**expired 2025**): `https://patents.google.com/patent/US7324663B2/en`
- US7497780 — integrated launch monitor UX: `https://patents.google.com/patent/US7497780B2/en`
- US7641565 — ball-placement detection: `https://patents.google.com/patent/US7641565B2/en`
- WAWGD (Foresight) filings index: `https://patents.justia.com/inventor/wawgd-inc-dba-foresight-sports`
- Foresight–Uneekor 2024 license (Business Wire): `https://www.businesswire.com/news/home/20240930838753/en/`

Acushnet / Titleist (Gobush et al.)

- US5501463 — stereo + retroreflective dots, club and face (**expired**): `https://patents.google.com/patent/US5501463A/en`
- US6500073 — stereo ball trajectory + flight integration (**expired**): `https://patents.google.com/patent/US6500073B1/en`
- US6758759 — dual stereo monitors, measured face angle (**expired**): `https://patents.google.com/patent/US6758759B2/en`
- US7143639 — portable four-camera monitor (**expired**): `https://patents.google.com/patent/US7143639B2/en`
- US8500568 / US8556267 — portable continuations: `https://patents.google.com/patent/US8500568B2/en`, `https://patents.google.com/patent/US8556267B2/en`
- US10668350 — 3D/light-field launch monitor: `https://patents.google.com/patent/US10668350B2/en`
- US6186002 — C_D/C_L from measured trajectories (model calibration template): `https://patents.google.com/patent/US6186002B1/en`

Others

- Full Swing US2020/0147470 — CW+FMCW dual-mode launch monitor: `https://www.freepatentsonline.com/y2020/0147470.html`
- Creatz US10247553 — start-position sensor: `https://patents.google.com/patent/US10247553B2/en`
- Golfzon US9242158 — two-stage latency-hiding simulation: `https://patents.google.com/patent/US9242158B2/en`
- Uneekor patent/license list: `https://uneekor.com/legal/patents`
- Camera-timing/Gabor dimple tracking US12401909: `https://patents.google.com/patent/US12401909B2/en`
- TrackMan EP1698380 litigation coverage: `https://thegolfwire.com/322771-2/`
- FlightScope 2022 BGH win vs TrackMan: `https://golfbusinessnews.com/news/innovation-centre/flightscope-wins-patent-infringement-case-against-trackman-in-germany/`

A.2 Regulatory / teardown (RF ground truth)

- TrackMan 4 FCC ID SFX-TMAN4 (X-band + 24 GHz): `https://fccid.io/SFX-TMAN4`
- TrackMan iO FCC ID SFX-TMB0201 (24 GHz): `https://fccid.io/SFX-TMB0201`
- FlightScope Mevo FCC ID QXP-A7310 (24.125 GHz): `https://fccid.io/QXP-A7310`
- Mevo+ teardown / QXP-LJ361 (10.5–10.55 GHz) discussion: `https://golfsimulatorforum.com/forum/flightscope/277362-mevo-teardown`

A.3 Manufacturer technical documentation

TrackMan

- 40+ parameters explained: `https://www.trackman.com/blog/golf/40-trackman-parameters`
- Club data definitions: `https://www.trackman.com/blog/golf/club-data-definitions`
- Club speed (geometric-center definition): `https://www.trackman.com/blog/golf/what-is-club-speed`
- Spin loft / spin rate: `https://www.trackman.com/blog/golf/spin-loft`, `https://www.trackman.com/blog/golf/spin-rate`
- OERT ("Two radars, one camera, zero doubt"): `https://www.trackman.com/blog/golf/two-radars-one-camera-ze`
- Tech specs (TM4, iO): `https://www.trackman.com/golf/launch-monitors/tech-specs`
- Ball Flight Laws newsletter (Jan. 2009; D-plane, and the origin of the "85/15" rule of thumb—see §3.1.1 for why the measured horizontal weight is 0.76): `https://www.yumpu.com/en/document/view/11652756/trackman-ball-flight-laws`

A.3.1 The TrackMan newsletter archive

Every official TrackMan newsletter URL now returns 404, and the Wayback captures are redirect stubs. A complete live third-party mirror of issues #1–#10 exists and is the richest single source of TrackMan-published numbers anywhere. All at `https://www.gregsmithgolfcoach.com/wp-content/uploads/2014/04/`:

- `newsletter1.pdf` (Nov 2007) through `newsletter9.pdf` (Jan 2013), sequentially numbered.
- `TrackMan_Newsletter_2014.pdf` is issue #10 (Jan 2014); the filename convention changed, which is why sequential guesses fail. The series ends there.
- **#7 (Oct 2010), "Ten Fundamentals"** is the origin of most TrackMan rules of thumb, including the 85/15 claim in both planes, the spin-loft subtraction stated as a definition, and "the spin rate drops during ball flight—typically 4% for each second."
- **#9 (Jan 2013)** carries the canonical club-delivery definitions (all specifying maximum compression) and the only published TrackMan accuracy specification located: for TrackMan III/IIIe at 95% confidence, club speed ±1.5 mph, attack angle ±1.0°, club path ±1.0°, dynamic loft ±0.8°, face angle ±0.6°.
- **#8 (Jun 2011)** quantifies bulge: a 12.7 mm heel-ward impact makes the face angle 2° closed at that point relative to face centre, "for all drivers on the market."

Caution

Those TrackMan accuracy figures are the manufacturer's own. Independent testing against a traceable optical benchmark (Leach et al. 2017, `https://doi.org/10.1016/j.measurement.2017.08.009`) found club path met ±1° on only 45% of shots and face angle on 66%, and that full clubhead parameters were returned on just 62% of shots overall—19% for a utility wedge. For OpenFlight the lesson is that a published tolerance and a per-shot success rate are different specifications, and only the first is ever advertised.

Foresight Sports

- GCQuad product/technology: `https://www.foresightsports.com/products/gcquad-launch-monitor`
- Ball and club data (spherical correlation, fiducials): `https://foresightsports.eu/ball-club-data/`
- HMT head measurement (dot-count data tiers): `https://www.foresightsports.com/hmt-head-measurement`
- Club marker application guide: `https://help.foresightsports.com/hc/en-us/articles/4408197030035`
- Marker troubleshooting: `https://help.foresightsports.com/hc/en-us/articles/4405916455443`
- Design history (Popular Science): `https://www.popsci.com/gear/foresight-sports-quadmax-photometric-golf-`

Other vendors

- FlightScope X3 (multi-frequency, Fusion Tracking): `https://thegolfwire.com/326713-2/`
- Garmin R10 architecture and accuracy: `https://mygolfsimulator.com/garmin-r10-data/`, `https://mygolfsimulator.com/garmin-r10-accuracy/`
- Garmin R50 (three-camera): `https://www.garmin.com/en-US/p/736810/`
- Full Swing KIT technology and specs: `https://www.fullswinggolf.com/kit-launch-monitor-technology/`, `https://www.fullswinggolf.com/kit-specs/`
- Rapsodo MLM2PRO spin (MyGolfSpy): `https://mygolfspy.com/news-opinion/rapsodo-mlm2pro-takes-spin-mea`
- SkyTrak+ measurement: `https://www.skytrakgolf.com/pages/what-does-the-skytrak-st-launch-monitor-mea`
- Uneekor EYE XO / Dimple Optix: `https://uneekor.com/blogs/blog/photometric-vs.-doppler:-which-launch-monitor-technology-delivers-the-most-accurate-golf-data`
- Uneekor calibration guide (Carl's Place): `https://www.carlofet.com/blog/calibrating-uneekor-launch-monitors` official PDF: `https://download.uneekor.com/docs/EYEXO2_Calibration_Guide.pdf`
- ProTee VX reviews: `https://www.playbetter.com/blogs/golf-simulator-reviews/protee-vx-review`

A.4 Peer-reviewed and academic

- Penner, "The physics of golf," Rep. Prog. Phys. 66 (2003): `https://iopscience.iop.org/article/10.1088/0034-4885/66/2/202`
- Leach et al. 2017 launch-monitor validation (*Measurement*): `https://www.sciencedirect.com/science/article/abs/pii/S0263224117305079`; open-access: `https://repository.lboro.ac.uk/articles/journal_contribution/9562799`

- TrackMan 4 indoor reliability (J. Sports Sci. 2024): https://www.tandfonline.com/doi/full/10.1080/02640414.2024.2314864
- Mevo+ vs TrackMan 4 agreement (2025): https://www.sciencedirect.com/science/article/pii/S2772696725000420
- Bearman & Harvey 1976 golf ball aerodynamics: https://www.cambridge.org/core/journals/aeronautical-quarterly/article/abs/golf-ball-aerodynamics/67FE0903DB1CC12001F1ED1B1261C4B9
- Smits & Smith aerodynamic model: https://www.researchgate.net/publication/284037213
- USGA Indoor Test Range conditions: https://www.usga.org/content/dam/usga/images/equipment-standards/ITR-test-conditions-2028-ODS.pdf
- Friction/tangential compliance and launch angle (MDPI 2020): https://doi.org/10.3390/proceedings2020049027
- Face angle / club path influence study: https://www.researchgate.net/publication/323373897
- Cross oblique-impact golf experiments: https://www.physics.usyd.edu.au/~cross/GOLF/GOLF.htm
- Cross & Nathan oblique collision formalism: https://arxiv.org/pdf/1610.03464
- Dynamic models review, Sports Engineering 25 (2022): https://link.springer.com/article/10.1007/s12283-022-00387-0
- Stanford LES of golf ball flow (Sports Eng. 2019): http://aero-comlab.stanford.edu/Papers/golf_ball_sports_engineering_2019.pdf
- Golf ball aerodynamics in still air (MDPI): https://www.mdpi.com/2504-3900/2/6/238
- Micro-Doppler of spinning projectiles on CW radar: https://www.researchgate.net/publication/317119893
- SpinDOE camera spin estimation (transferable method): https://arxiv.org/pdf/2303.03879
- "Measuring Ball Spin by Image Registration": https://www.researchgate.net/publication/2881136
- Doppler ball-tracking thesis (WSU, TrackMan platform): https://baseball.physics.illinois.edu/trackman/JasonMartinThesisWSU.pdf
- Alan Nathan's TrackMan (baseball) technical notes: https://baseball.physics.illinois.edu/trackman.html
- Cochran & Stobbs, *Search for the Perfect Swing* (1968): https://archive.org/details/searchforperfect0000coch

A.5 Engineering references (Tutelman et al.)

- 3D launch conditions from impact conditions: https://www.tutelman.com/golf/ballflight/3dlaunch.php
- Smash factor: https://www.tutelman.com/golf/ballflight/smashfactor.php
- Gear effect: https://www.tutelman.com/golf/ballflight/gearEffect1.php
- Spin decay: https://www.tutelman.com/golf/ballflight/spinDecay.php
- Ball flight laws summary: https://www.perfectgolfswingreview.net/ballflight.htm
- TrackMan face-angle derivation discussion (Manzella forum): https://forum.brianmanzellagolf.com/threads/how-does-trackman-flightscope-measure-the-club-face-angle.16591/

A.6 DIY / open-source

- PiTrac GitHub (GPL-2.0): `https://github.com/PiTracLM/PiTrac`
- PiTrac documentation: `https://docs.pitrac.org/`
- PiTrac Hackaday.io project (design logs): `https://hackaday.io/project/195042-pitrac-the-diy-golf-launch-mo`
- OpenFlight upstream: `https://github.com/jewbetcha/openflight`
- OpenFlight Grafana write-up: `https://medium.com/grafana-labs/openflight-building-an-open-source-golf-l`
- GC2-imitating OpenCV experiments: `https://github.com/ronheywood/opencv`
- GSA Golf simulator theory (marked-ball DIY methods): `https://www.golf-simulators.com/GolfSimulatorTheory.html`

A.7 Hardware datasheets, protocols, and standards (OpenFlight stack)

- OPS243 product brief: `https://omnipresense.com/wp-content/uploads/2024/01/OPS243-Product-Brief_004-F.pdf`
- OPS243 AN-010 API interface: `https://omnipresense.com/wp-content/uploads/2025/10/AN-010-AD_API_Interface.pdf`
- OPS243 AN-027 rolling buffer: `https://omnipresense.com/wp-content/uploads/2025/06/AN-027-A_Rolling-Buffer-1.pdf`
- OPS243 AN-029 sports applications (cites OpenFlight): `https://omnipresense.com/wp-content/uploads/2026/06/AN-029-A_OPS243-for-Sports_260621.pdf`
- RFbeam K-LD7 datasheet: `https://rfbeam.ch/product/k-ld7-radar-transceiver/`
- TI IWR6843 datasheet: `https://www.ti.com/lit/ds/symlink/iwr6843.pdf`; ISK EVM: `https://www.ti.com/tool/IWR6843ISK`; placement app note: `https://www.ti.com/document-viewer/lit/html/SWRA758`
- TI mmWave golf-swing forum thread: `https://e2e.ti.com/support/sensors-group/sensors/f/sensors-forum/1116424/iwr6843aop-golf-swing-analyzer-with-mmwave-radar`
- Raspberry Pi camera documentation (Global Shutter, XTR trigger): `https://www.raspberrypi.com/documentation/accessories/camera.html`
- GSPro Open Connect v1: `https://gsprogolf.com/GSProConnectV1.html`; community feedback: `https://github.com/tnbozman/gspro-interface/blob/main/OpenAPI-Documentation-Feedback.MD`; reference client: `https://github.com/travislang/gspro-garmin-connect-v2`
- USGA equipment standards: initial velocity TPX3007: `https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3007-initial-velocity-test-procedure.pdf`; ODS TPX3006: `https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3006-overall-distance-and-symmetry-test-procedure.pdf`; clubhead CT TPX3004 and MOI TPX3005 via `https://www.usga.org/equipment-standards.html`; 2028 ALC revision: `https://www.usga.org/content/usga/home-page/articles/2023/12/revised-golf-ball-testing-conditions-to-take-effect-in-2028.html`

- CFAR tutorial (CA/GO/SO/OS): `https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfa` `html`; Purdue lecture notes: `https://engineering.purdue.edu/~mrb/resources/AltLectureF/Session_21.pdf`
- Stalker baseball-spin patent (radar spin outside golf): `https://patents.google.com/patent/US10935657B2/en`
- Weibel Scientific projectile-tracking radar (TrackMan's origin): `https://patents.google.com/patent/EP1735637B1/en`

A.8 Comparative testing and market analysis

- MyGolfSpy indoor accuracy test: `https://mygolfspy.com/news-opinion/three-of-the-most-accurate-indoor-la`
- GCQuad vs TrackMan robot spin data: `https://golfsimulatorzone.com/gcquad-vs-trackman-accuracy-in-2026/`
- Foresight GCQuad/GC3 comparison: `https://www.foresightsports.com/blogs/golf-tips/comparing-golf-laun`
- Bushnell Launch Pro vs GC3 vs GCQuad: `https://www.playbetter.com/blogs/golf/bushnell-launch-pro-vs-gc3`
- Photometric vs radar explainers: `https://www.playbetter.com/blogs/golf-simulator-comparisons/photometric-vs-radar-golf-launch-monitor`, `https://golfbays.de/en/blogs/news/radar-vs-photometric-la`
- Indoor flight compensation: `https://www.cerogolf.com/post/how-launch-monitors-compensate-for-indoor-bal`
- TrackMan history (Golf Monthly, Tuxen profile): `https://www.golfmonthly.com/features/from-tracking-missile`

Appendix B

Detailed Implementation Guidance for OpenFlight

This appendix turns the review into engineering guidance at the level of signal-processing parameters, algorithms, and procedures, citing the governing sources throughout. Coordinate conventions follow Chapter 2.

B.1 Radar signal chain (OPS243-A)

B.1.1 Doppler scaling and resolution budget

At $f_c = 24.125\,\text{GHz}$, Equation (4.1) gives 71.7 Hz per mph. With OpenFlight's 30 kHz I/Q sample rate, the unambiguous span is $\pm 15\,\text{kHz} \approx \pm 209\,\text{mph}$ — adequate for ball speeds to $\sim$200 mph. Velocity resolution is set by observation time: a 128-sample window (4.27 ms) gives $\Delta f = 234\,\text{Hz}$ $\approx 3.3\,\text{mph}$ per raw bin; zero-padding to 4096 (the current pipeline) interpolates the peak but does not add information. For *spin* work the window, not the padding, must grow: resolving sidebands at $f_{\text{spin}} = S/60$ (e.g. 50 Hz at 3,000 rpm) requires windows of $\gtrsim 40\,\text{ms}$, during which a 150 mph ball decelerates and the central line *chirps* — so the practical estimator de-chirps first (track the central line per US8845442's spectral-trace step [71], resample the phase to remove it), then measures the residual modulation.

B.1.2 Spin-rate comb estimation

The patent-documented chain (Section 4.4.2) maps to this concrete pipeline [71, 62]:

1. STFT with 50–75% overlap over the post-impact 50–250 ms; track the ball ridge (max-SNR bin per frame with continuity constraint).
2. De-chirp: mix each frame down by the tracked ridge frequency so the ball line sits at DC.
3. Estimate the sideband comb spacing on the de-chirped spectrum via cepstrum or harmonic product spectrum — both are robust to missing harmonics; the patent's *qualification* step (verify equal spacing across ≥ 3 consecutive frames, and consistency of the harmonic-number assignment) is what separates real spin from clutter [71].
4. Report spin only when the qualified comb persists; otherwise fall back to a flagged estimate from club priors (Section B.3) — the Garmin R10's documented behavior [35].

Expected detection physics: sideband amplitude scales with ball-surface asymmetry [71]; range balls and clean urethane balls read weakly (the reason for Titleist RCT metal-tagged balls [37] and FlightScope's metallic-dot stickers [19]). Logging which ball type produced each detection will quantify this for OpenFlight's own statistics.

Patent posture: this method is claimed by US8845442 to ∼2029 (US10393870 to Dec. 2026) [71]; the FlightScope phase-demodulation alternative (US9868044, dielectric lens [27]) is claimed to ∼2034. See Section 7.6.

B.1.3 Club-speed extraction

The pre-impact spectrum contains a velocity *smear* from heel (slow) to toe (fast), plus shaft returns below. TrackMan resolves this by reconstructing the head silhouette and reporting the geometric center [67]; a single-radar approximation that tracks a stable reference is: gate the last 30–50 ms before the trigger timestamp, form the velocity histogram of CFAR-passing bins, discard the bottom decile (shaft/hosel) and top decile (toe glint), and report a fixed percentile (median of the remainder). Validate against the loft-dependent smash ceiling of Section 3.2 ([69]); reject or flag shots exceeding it, which independent testing shows is the signature of toe-lock errors [31].

B.2 Angle measurement (K-LD7) and the EKF

B.2.1 Interferometric angles

The K-LD7's two receive patches implement Equation (4.2) with a $\lambda/2$-class baseline; per-bin phase comparison of the two ADC channels after the range-Doppler FFT yields one angle per detection — the same phase-monopulse principle as US10850179 [72] and the R10's three-receiver array [36]. Two practical constraints from the commercial art: (i) angle error grows as SNR falls, so weight each detection by measured SNR; (ii) mechanical alignment dominates the error budget — a 1° mount error is a 1° bias on every launch direction, amplified ∼1.15× into face angle (Equation (4.6)).

B.2.2 Recommended filter

Implement the Section 8.3 smoother concretely as: state $\boldsymbol{x} = (\boldsymbol{p}, \boldsymbol{v})$ (add spin states later); process model Equation (8.1) with Smits–Smith coefficients [52]; measurements: OPS243-A radial speed ($\dot{r}$, high rate, low noise), K-LD7 angle+range detections (lower rate, SNR-weighted covariance). Run a forward EKF then an RTS (Rauch–Tung–Striebel) backward smoother over the 50–150 ms burst, and evaluate the smoothed state at the sound-trigger timestamp minus the acoustic delay — this back-extrapolation is how commercial radars report launch conditions robustly despite early-flight clutter [71, 25]. The innovation-gating step of the EKF doubles as the outlier rejector for multipath and club returns. The same filtering architecture extends from point tracking (the ball) to rigid-body tracking (the club) via the screw-theoretic formulation of Chapter E, which is the recommended estimation layer for the IWR6843 integration.

B.2.3 Alignment calibration procedure

Adopt the commercial patterns: (i) Uneekor-style floor chart — a printed target-line chart at known positions establishes the target-line azimuth for both K-LD7s [4]; (ii) FlightScope-style known-trajectory check — roll or swing balls along a surveyed line and verify reported directions (US10338209 uses a Doppler simulator for the same purpose [25]); (iii) record the mount pose in the session log so data from different setups is never silently mixed.

B.3 Derived club data, with priors

Implement the D-plane module once, use it both ways (Sections 3.1 and 4.5.2):

- **Forward** (simulation/validation): Tutelman's calibrated closed form, Equations (3.3) and (3.4), plus spin from Equation (3.7) [70].
- **Inverse** (reporting): face angle from Equation (4.6) with w_f interpolated in dynamic loft between 0.87 (driver) and 0.75 (wedge) [59, 70]; dynamic loft analogously from launch angle and attack angle. Tag both as *derived* per Table 2.3.
- **Priors table** per club type (driver. . . wedge): static loft, typical dynamic-loft delta, spin-loft range, smash ceiling (Section 3.2), PGA/LPGA reference deliveries [65]. Used for: spin fallback estimates, outlier gating, and the smash sanity check.
- **Gear-effect bound**: when impact location is unknown, attach an uncertainty of up to $\pm 6°$ of spin-axis tilt per 0.14 in of possible driver miss [44, 68] to any derived face-to-path interpretation — and surface it in the UI rather than hiding it.

B.4 Optical module (Phase 2) design parameters

The expired Wintriss patents [29] plus PiTrac's published design [45] fix the working parameters:

- **Sensor**: Raspberry Pi Global Shutter camera (IMX296, 1456×1088); global shutter is non-negotiable (rolling shutter skews a 150 mph ball by several pixels per row-time).
- **Strobed multi-exposure**: $N = 3$–5 IR pulses (850 nm, tens of µs each) inside one long exposure freeze N ball images per frame; at 150 mph the ball moves 6.7 cm/ms, so pulse spacing of $\sim$300–500 µs spaces images 2–3 cm apart in a 30 cm capture volume — matching the commercial capture geometry [11, 45].
- **Trigger**: reuse the SEN-14262 sound trigger; its $\sim$10 µs latency is far inside the strobe-timing budget, solving the triggering problem the Wintriss patent addressed with a microphone+radar pair [29].
- **Ball detection**: Hough circles on the strobed frame (PiTrac-proven [45]); depth from the known 42.67 mm diameter (the Wintriss mono-camera range cue [29]) or from a second camera via Equation (5.1).
- **Spin**: register the dimple texture between successive ball images over $SO(3)$ (Equation (5.2)); Gabor-filter pre-enhancement of dimples is documented in US12401909 [58]; with $\Delta t \approx 400$ µs, 3,000 rpm is only 7.2° of rotation — comfortably below the aliasing limit, and small enough that a local search around the D-plane-predicted axis converges quickly (Section 5.2.2). This is the public-domain (post-2025) Wintriss/Foresight method [29, 10].
- **Impact location / measured face angle** (later): either face fiducials per the expired Acushnet recipes (US5501463/US6758759: dots + stereo pose) [17, 30], or an overhead second camera per the Uneekor geometry [73].

B.5 Flight model and validation protocol

Ship Smits–Smith (Equation (8.2)) with spin decay [52], version the coefficients, and calibrate against measured trajectories using the US6186002 fitting approach [33] — outdoor sessions with the MLM2PRO (or simple carry ground-truth) provide the data. Validation against the MLM2PRO should follow the Leach protocol structure [31]: per-club shot blocks, Bland–Altman limits of agreement per parameter, ball data compared directly, spin compared only on RPT/RCT balls (where the MLM2PRO's spin is camera-measured [37]), and club speed compared with an explicit reference-point caveat (Section 4.5). Log raw I/Q and (later) raw frames for every shot so algorithm changes can be replayed against history — the practice that made this review's accuracy analysis possible for the commercial units is exactly what an open project can do better.

Appendix C

Sensor Hardware and Integration Reference

This appendix is a specification-level reference for OpenFlight's actual and planned building blocks, drawn from vendor datasheets and application notes (all linked). Notably, OmniPreSense's own sports application note AN-029 documents the exact OpenFlight golf configuration and cites the OpenFlight repository by name as its reference implementation [39].

C.1 OmniPreSense OPS243-A (24 GHz CW Doppler)

C.1.1 Hardware

Per the product brief [42]: 24.00–24.25 GHz ISM band, 11 dBm transmit power (FCC ID 2ALLL243A), patch antenna with **20° azimuth × 24° elevation** −3 dB beamwidth (footprint ∼0.4 m wide at 1 m, 1.8 m at 5 m); motion detection 1–100 m; speed to 348 mph at 50 ksps; accuracy spec 0.5%; USB CDC + 3.3 V UART (default 19,200 8N1); 5–24 V supply, 1.7 W active.

C.1.2 API essentials

From AN-010 [40]: sample rate `S=n` (1–1000 ksps; 10 ksps default); buffer 1024/512/256/128 via `S>`/`S<`/`S[`/`S(`; zero-padding `Xn`/`X=16`/`X=32` to a 4096-point FFT. Speed ceiling and resolution scale with sample rate (10 ksps → 31.1 m/s at 0.061 m/s; 50 ksps → 155.4 m/s at 0.304 m/s per 1024-sample buffer). Output modes: `OJ` JSON, `OT` timestamps, `OM` magnitudes, `O=n` multi-object (to 16), `OF` post-FFT, `OR` raw I/Q. Filters: `R>`/`R<` speed, `R`± direction, `M>` magnitude, `K+` peak averaging; built-in cosine-error correction `^`/±`n.n` (0–89°). `A!` persists settings to flash.

C.1.3 Rolling buffer and triggering

From AN-027 [41]: `G1` enters rolling-buffer mode with a fixed **4096-sample I/Q buffer organized as 32 segments of 128 samples**; trigger by software (`S!`) or a 3.3 V rising edge on **J3 pin 3 (HOST_INT)**; `S#n` sets the pre/post-trigger split (default 8 → 1024 pre + 3072 post). At OpenFlight's 30 ksps the buffer spans 136.5 ms with a 208.5 mph ceiling — AN-027's "golf ball setting." The app note wires a **SparkFun SEN-14262 Gate output directly to HOST_INT** (OpenFlight's exact trigger) and flags the acoustic-latency budget: sound from 2 m arrives ∼6.6 ms late, so the pre-trigger split must cover the impact-to-trigger gap (their example `S#18`).

C.1.4 The vendor golf recipe (AN-029)

AN-029 [39] specifies: `S=30` (209 mph ceiling), `S(` 128-sample segments, `X=32` (4096-point FFT, 0.1 mph resolution, ∼200 Hz report rate), `US`, `R>10` to mask waggle, `M>10`, `O2` to report ball + club for smash factor (gated 1.0–1.50, matching Section 3.2), `K+` averaging, sensor 2–3 m behind the ball. Golf balls are rated "high" reflectivity, detectable 5–10 m.

C.2 RFbeam K-LD7 (24 GHz FSK, dual-RX angle)

Per the datasheet [50]: 24.050–24.250 GHz FSK (two frequencies, enabling range via phase difference); EIRP 6 dBm; 1 TX + **2 I/Q RX patches at 6.223 mm ($\approx \lambda/2$) spacing** — the interferometric baseline of Equation (4.2); beam 80° H × 34° V. Per-frame 256-point complex FFT; **angle ±90° at 1° resolution from the Rx1–Rx2 phase difference**; distance 5 cm–100 m (resolution 5 cm at the 5 m range setting); speed 0.1–100 km/h — note the **62 mph speed ceiling**, which confines the K-LD7 to angle and club-speed work; ball speed must come from the OPS243-A. Frame time at the 100 km/h setting is 29 ms (∼34 Hz).

UART protocol (115200 8E1 default, to 3 Mbaud via `INIT`): message types `RADC` (raw ADC: 256 I + 256 Q for Rx1@f_A, Rx2@f_A, Rx1@f_B — the payload OpenFlight's interferometry consumes), `RFFT` (spectrum + threshold), `PDAT` (up to 12 raw targets: distance/speed/angle/magnitude), `TDAT` (tracked target), `DDAT` (flags), `DONE` (frame counter — use it to detect dropped frames); requested via the `GNFD` bitfield. Configuration: `RSPI` max speed, `RRAI` max range, `THOF` threshold offset (10–60 dB), `RBFR` base frequency (three channels for multi-module coexistence — set the two OpenFlight units to different channels), `TRFT` tracking filter, detection-window bounds (`MIRA`/`MARA`/`MIAN`/`MAAN`/`MISP`/`MASP`). Positive speed = receding.

C.3 TI IWR6843 (60–64 GHz FMCW MIMO, on order)

Per the datasheet [54]: 60–64 GHz with **4 GHz chirp bandwidth** (∼3.75 cm native range resolution); **3 TX / 4 RX = 12 virtual antennas** (TDM-MIMO); on-chip C674x DSP + radar hardware accelerator (FFT, log-magnitude, CFAR); complex-baseband ADC to 12.5 Msps. The LEVM/ISK-class EVMs give ∼120° azimuth FoV with ∼15° azimuth angular resolution (8 virtual azimuth antennas) and coarse elevation; the AOP variant trades resolution for a 130°×130° field. The out-of-box mmWave SDK demo streams a TLV point cloud (x, y, z, Doppler) at ∼10–20 Hz — *too slow for a 3 ms launch window*; using the IWR6843 for launch measurement requires a custom chirp/frame configuration (short frames, high Doppler span) and low-level processing, for which TI's people-tracking labs (TIDEP-01000/01010) are the closest starting points [54]. TI's forums confirm no golf-specific lab exists. The natural OpenFlight role: a single sensor that measures range, angle, and radial speed simultaneously (replacing both K-LD7s) once custom chirp work is done.

C.4 Raspberry Pi Global Shutter camera (optical module)

Sony IMX296 sensor: 1456×1088, 3.45 µm pixels, true global shutter, exposures to ∼30 µs, max ∼60 fps streaming [49]. The key feature for Section B.4 is the **XTR external-trigger pad**: pulse low to expose (exposure = pulse width + 14.26 µs); frame rate follows the pulse train; multiple cameras on one trigger line are hardware-synchronized; enable with `v4l2-ctl -c trigger_mode=1` (early boards: remove R11 if Q2 is fitted) [49]. Continuous 60 fps cannot capture flight — the strobed multi-exposure design of Section B.4 is the correct use of this sensor.

C.5 Simulator integration: GSPro Open Connect

GSPro's Open Connect v1 [22] is the only fully documented open launch-monitor protocol and should be OpenFlight's native output (PiTrac ships it; E6/TruGolf and Foresight FSX require partnership agreements [45]). Mechanics: JSON over TCP to `127.0.0.1:0921`, launch monitor as client. Minimum shot message: `DeviceID`, `ShotNumber`, `APIversion`, `ShotDataOptions{ContainsBallData, ContainsClubData}`, and `BallData` with the five required fields:

Field	Units	Required	Maps to
`Speed`	mph	yes	ball speed (Table 2.2)
`VLA`	deg	yes	launch angle
`HLA`	deg	yes	launch direction
`TotalSpin`	rpm	yes*	spin rate
`SpinAxis`	deg	yes*	spin-axis tilt (− = draw)

*or `BackSpin`+`SideSpin`, related by Equation (2.2).

Optional `ClubData` carries `Speed`, `AngleOfAttack`, `Path`, `FaceToTarget`, `Loft`, `Lie`, `SpeedAtImpact`, `VerticalFaceImpact`, `HorizontalFaceImpact`, `ClosureRate` — precisely the Table 2.1 set, so the provenance tagging of Chapter 10 carries through unchanged. Status flags (`LaunchMonitorIsReady`, `LaunchMonitorBallDetected`, `IsHeartBeat`) and the 201 player-info response (handedness, selected club — useful for per-club priors, Section B.3) complete the loop.

C.6 Regulatory constants for the physics engine

USGA/R&A equipment rules anchor the models of Chapters 3 and 8 [76]: ball mass ≤ 45.93 g and diameter ≥ 42.67 mm (the constants in Equation (8.1)); clubhead characteristic time CT $\leq 239\,\mu$s (+18 tolerance) $\approx$ COR 0.830 — the e in Equation (3.6); head MOI $\leq 5900\,\mathrm{g\,cm}^2$ (+100) about the vertical CG axis — the upper bound on I_h in Equation (3.8); volume ≤ 460 cc. The Overall Distance Standard (317 yd + 3 at 120 mph clubhead / 10° / 2520 rpm; from January 2028, 125 mph / 11° / 2200 rpm) is a useful end-to-end sanity envelope for the flight model: a conforming ball simulated at ALC conditions must not materially exceed 320 yd [76].

C.7 Detection theory: CFAR selection

OpenFlight's current CFAR (SNR > 15 over a 150-bin DC mask) is a cell-averaging scheme. The radar literature [32] distinguishes: CA-CFAR (mean of training cells around the cell under test, guard cells excluded; threshold $\alpha\hat{P}_n$ with α set by the desired false-alarm probability) — optimal in homogeneous noise; and **OS-CFAR** (order statistic: the k-th ranked training cell) — robust when two targets sit close together, at ∼0.5–1 dB detection loss. The club-then-ball geometry of a golf shot is precisely the two-closely-spaced-targets case, so OS-CFAR is the better default for the impact window.

Appendix D

Patent Portfolio Compendium

This appendix enumerates, company by company, every US patent identified in the portfolio sweep (July 2026), with each number hyperlinked to its Google Patents page. Coverage notes: all 45 numbers on TrackMan's official legal page [63] are included plus seven granted TrackMan patents absent from that page; Uneekor's marking page [74] and Justia/FreePatentsOnline assignee sweeps were used as completeness cross-checks. Priority years are US/PCT filing-based (Korean assignees typically claim a KR priority ∼12 months earlier). Status is as reported by Google Patents; *verify claim-by-claim with counsel before relying on any entry.*

Two attribution corrections surfaced by this sweep are worth flagging prominently: (i) the widely cited "radar + image data 3D tracking" family US10596416 / US11697046 / US12128275 belongs to **Topgolf Sweden AB (Toptracer)**, not TrackMan; and (ii) Full Swing's launch-monitor application US2020/0147470 granted as US11311789B2 (expiry ∼2039).

D.1 TrackMan A/S (incl. Interactive Sports Games A/S)

Radar fundamentals and target-line deviation (2004–2011)

Patent	Subject	Prio.	Notes
US8085188B2	Deviation of launched projectile vs. image-designated target direction	2004	family expires 2026–27
US8912945B2	Continuation: camera+radar launch/target/ trajectory correlation	2004	not on legal page
US9857459B2	Continuation: camera on radar identifies target feature	2004	lapsed 2022
US10473778B2	Continuation	2004	
US10690764B2	Continuation	2004	
US9958527B2	Direction-of-arrival sensor: extra RX antenna resolves monopulse phase ambiguity	2011	the sparse-array geometry of Section 4.2

Spin rate and spin axis

Patent	Subject	Prio.	Notes
US8845442B2	Spin rate via harmonic sidebands; spin axis via trajectory/Magnus inversion	2005	to ∼2029 (PTA); EP 1 698 380 litigated
US9645235B2	Continuation	2005	
US10393870B2	Continuation	2005	to Dec. 2026
US10962635B2	Continuation	2005	not on legal page
US11143754B2	Continuation	2005	not on legal page
US10850179B2	Spin axis from multi-receiver Doppler decomposition	2018	
US11446546B2	Continuation: phase differences → axis	2018	
US11938375B2	Continuation: ≥3 non-colinear receivers	2018	
US11673029B2	Marked-ball radar spin (great-circle marker layout)	2019	the RCT-ball patent
US12179068B2	Continuation	2019	
US12042698B2	Toppling frequency of non-spherical rotating objects	2018	

Club impact (markerless, single camera)

Patent	Subject	Prio.	Notes
US10953303B2	Impact when/where via fixed club points across frames	2017	the OERT impact-location family
US11439886B2	Single camera, no markers or stereo	2017	
US11612801B2	Continuation	2017	
US12263393B2	Fix points + fix lines → 3D orientation	2017	

Radar+camera fusion, calibration, tracer, range, short game

Patent	Subject	Prio.	Notes
US10052542B2	Coordinating radar + image data; overlay	2004	
US10471328B2	Continuation	2004	
US10989791B2	Fused radar range/rate + imager angles track	2016	core fusion patent
US11828867B2	Continuation	2016	
US11619708B2	Inter-sensor calibration by track comparison	2020	
US12517218B2	Continuation: automatic calibration	2020	
US11748985B2	Master clock; composite multi-moment images	2019	

Patent	Subject	Prio.	Notes
US12067775B2	Continuation	2019	
US12586211B2	Imager event detection; camera power-mode switch	2023	not on legal page
US9855481B2	Broadcast tracer overlay	2009	5-patent family: also US10315093B2, US10441863B2, US11135495B2, US11291902B2
US10379214B2	Multi-bay range tracking (one radar, many bays)	2016	also US11086005B2, US11921190B2, US12618962B2
US11452911B2	Bay imager + range radar arbitration	2019	also US11986698B2
US12036465B2	Player ID via wearable + trajectory correlation	2021	
US12186643B2	Camera line-of-sight ∩ terrain model → ball rest position	2021	
US10444339B2	Bounce/slide/roll classification from velocity profile	2016	also US11079483B2, US11619731B2, US11946997B2 (green speed)
US11285367B2	Strategy simulation from player capability	2018	also US12109473B2
US11951372B2	Mishit filtering, optimal-shot analytics	2020	also US12539454B2
US12616891B2	Automated ball/strike, biometric strike zone	2021	baseball; not on legal page

D.2 Topgolf Sweden AB (Toptracer / Protracer)

A separate company from TrackMan; camera-first range tracking.

Patent	Subject	Prio.	Notes
US8077917B2	Enhancing images in sports video (the founding Protracer tracer, Forsgren)	2006	
US10596416B2	3D tracking: radar + image data	2017	also US11697046B2, US12128275B2
US10898757B2	3D tracking: radar speed + 2D image	2020	also US11504582B2, US11883716B2, US12330020B2

Patent	Subject	Prio.	Notes
US11335013B2	Motion-based pre-processing, virtual time sync	2020	also US11557044B2, US12322122B2
US11644562B2	Trajectory extrapolation, origin determination	2020	also US11771957B2, US12121771B2, US11964188B2
US11513208B2	Camera-based projectile spin	2021	also US12105184B2; club-parameter spin US12544624B2
US11995846B2	Tracking with unverified detections	2021	also US12361570B2
US11815618B2	Doppler radar coexistence	2021	also US12253622B2
US12206977B2	Predictive camera control	2022	
US12298326B2	Wind velocity estimation	2022	
US12594460B2	Blob management for projectile tracking	2023	

D.3 FlightScope / EDH (Henri Johnson)

Patent	Subject	Prio.	Notes
WO2003032006A1	Foundational golf-ball tracking: Doppler + antenna array phase monopulse	2001	GB/WO only; expired art
US8189857B2	Bounce-mark detection + tracking (cricket)	2007	
US9036864B2	Trajectory and bounce position	2011	reinstated
US9868044B2	Spin rate from phase modulation (dielectric-lens)	2013	to ∼2034; reinstated
US10775492B2	Spin axis from perpendicular receiver pairs	2013	to ∼2035
US10338209B2	Fusion Tracking (multi-receiver + camera)	2015	also US11016188B2
US11573082B2	Tracking in varied environmental conditions	2019	
US12528005B2	Weather-based range prediction, club selector	2023	
US20160306036A1	Putting-green tracking	2013	abandoned; citable art
US20180239012A1	Antenna with boresight optical system	2013	abandoned; Fusion hardware disclosure

D.4 Full Swing Golf

Patent	Subject	Prio.	Notes
US11311789B2	CW + FMCW dual-mode radar, non-uniform array (the KIT patent; grant of US2020/0147470)	2018	to ∼2039; also US11844990B2
US11875517B2	Frame-difference ball tracking, screen impact point	2020	also US12354282B2
US8758103B2	IR light-curtain translation + imaging rotation (legacy simulators)	2009	also US9616346B2, US11033826B2
US8926416B2	Simulator: spin via image analysis	2007	also US10058733B2
US8414408B2	Ball-permeable screen, ball return	2009	also US8834284B2

Caution: US10605910B2/US11086008B2 (Alphawave Golf) and US11565166B2 (individual) surface in "Full Swing" text searches but are unrelated assignees.

D.5 Garmin

US11351436B2 — "Hybrid golf launch monitor" (2019 priority): the Approach R10 patent, Doppler radar with camera supplement/correction. Garmin's golf-radar estate is essentially this single family; supporting art: US8647214B2 (2008, motion-sensor swing analysis), US7467060B2 family (2006, wearable motion-parameter estimation).

D.6 Rapsodo Pte. Ltd.

Patent	Subject	Prio.	Notes
US9955126B2	Core camera+radar moving-object analysis	2015	
US11170513B2	Marked-ball spin via surface-template matching	2016	the RPT-ball patent
US11747461B2	Radar+camera data fusion	2018	
US20210299540A1	3D reconstruction of the launch scene	2018	application
US20230065614A1	Spin detection/estimation pipeline	2021	application
US20230364468A1	Deep-learning ball/swing parameters from radar+image	2021	also club-side US20230070986A1
US12169941B1	Target-plane crossing localization	2024	
US12586248B2	Newest camera+radar fusion grants	2024	also US12548194B2
US12158517B1	Range-gated imager	2024	

D.7 Camera vendors: Foresight/Wintriss, Creatz/Uneekor, Golfzon

Foresight Sports (Wintriss → WAWGD → Wawgd Newco)

Patent	Subject	Prio.	Notes
US5333874A	IR light-curtain sports simulator (Kiraly/ Wintriss prehistory)	1992	expired
US7292711B2	Mono-camera photometric monitor; markerless dimple-feature spin	2002	**expired Apr. 2025**
US7324663B2	Smart-camera sibling	2002	**expired Aug. 2025**
US7497780B2	Integrated monitor UX (GC2 architecture)	2006	to 2027
US7540500B2	Foldable monitor housing	2006	to ∼2027
US7641565B2	Ball-placement detection / auto-arm	2006	to 2027
US8951138B2	Club head measurement (camera + optional inertial): face, path, loft/lie, impact	2012	the HMT/GCQuad club-data patent
US9737757B1	Alignment-stick target alignment	2016	
US10639537B2	Range tracking fused with launch monitor strike	2018	

Creatz Inc. (Uneekor)

Patent	Subject	Prio.	Notes
US9448067B2	Multi-camera (unsynchronized) trajectory via projection-plane intersection	2011	to 2033
US9605960B2	Single-camera plane-intersection trajectory	2011	to 2032
US9752875B2	Exposure/gain control from ambient brightness	2011	to 2032
US10247553B2	Start sensor: ball state at first movement	2011	to 2032
US10587797B2	Ball-image brightness compensation for spin marks	2016	to 2037
US10776929B2	Dynamic ROI from predicted ball motion	2016	to 2037
US11191998B2	Mark-based spin with model fallback	2018	to 2039
US12008770B2	Dimple-constellation markless spin (Dimple Optix)	2020	to 2042

Uneekor's marking page also lists the four licensed Wintriss/Foresight patents (US7497780, US7292711, US7641565, US7324663) [74, 3].

Golfzon Co., Ltd. (sensing core)

Patent	Subject	Prio.	Notes
US9242158B2	Two-stage sensing → simulation (latency hiding)	2011	to ∼2032
US9333409B2	Ball-candidate 2D-trajectory analysis, low-fps cameras	2011	also US9333412B2, US9162132B2
US9514379B2	Low-res launch + club trajectory → cheap spin estimate	2011	
US10045008B2	Unsynchronized stereo cross-acquisition (doubled frame rate)	2011	clever budget-hardware trick
US11364428B2	Spin fit by trajectory iteration vs. observed positions	2018	
US12002222B2	Database spin lookup with correction	2017	
US12599828B2	Marker-constellation spin between frames	2022	
US12605593B2	Rolling predictive ROI	2022	
US12478849B2	Single-camera putting-mat sensing	2021	
US20230347209A1	Stereo impact position on club face	2021	pending

D.8 Acushnet (Titleist) — the full lineage

Patent	Subject	Prio.	Notes
US4136387A	The ancestral optical impact/launch monitor	1977	expired
US5471383A	Shutterable cameras + retroreflective markers	1992	expired
US5501463A	Stereo + club/ball dots: face, path, contact location	1992	expired
US6241622B1	Portable dual-camera + strobe; aerodynamic trajectory	1998	expired; also US6533674B1 (multi-shutter), US6616543B1, US7086955B2
US6500073B1	Stereo trajectory + flight integration	1998	expired
US6758759B2	Dual stereo monitors; measured face angle	2001	expired
US7143639B2	Portable four-camera monitor	2004	expired; continuations US7395696B2 (optical fingerprinting, expired), US8500568B2, US8556267B2
US10668350B2	Stereo / light-field “true 3D” capture	2017	to ∼2038

Patent	Subject	Prio.	Notes
US6186002B1	C_D/C_L from measured trajectories	1998	calibration template

D.9 Others and foundational radar art

SkyTrak / SkyHawke: US12515116B2 (2023, swing-tag + launch-monitor fusion display) is the only relevant grant; the SkyTrak photometric unit itself is unmarked. **AccuSport, Zelocity, GolfTek, Ernest Sports, ProTee United:** no attributable US patents found — their methods rest on the expired Wintriss/Acushnet art. **Voice Caddie (Ucomm):** US10338212B2 (2014, portable Doppler swing/ball analyzer). **Sports Sensors (Dilz):** US6079269A (1997 Swing Speed Radar), US6898971B2, US8007367B2 (club speed + tempo) — all expired, free art. **Weibel Scientific** (TrackMan's engineering origin): EP1735637B1 (2004 multi-antenna CW Doppler tracking), MFCW ranging family (2014). **Applied Concepts (Stalker):** US10935657B2 (2019, baseball spin via Doppler micro-modulation) and the autocorrelation continuation — directly relevant prior art for radar spin outside golf.

Implication for OpenFlight

The compendium sharpens the FTO picture of Section 7.6: the *entire pre-2006 optical stack* (Acushnet 1977–2004, Wintriss 2002) and the *entire foundational radar-speed stack* (Sports Sensors, EDH 2001, Weibel 2004) are expired. What remains encumbered clusters in exactly three areas OpenFlight should treat carefully: radar spin extraction (TrackMan 2005 family to ∼2029; FlightScope 2013 families to ∼2034–35; Applied Concepts 2019), markerless camera club/impact measurement (TrackMan 2017 family; Foresight 2012 US8951138), and radar+camera fusion (TrackMan 2016/2020 families, FlightScope 2015, Rapsodo 2018–24, Topgolf Sweden 2017–23).

Appendix E

Clubhead Kinematics from Radar Velocities: A Screw-Theoretic How-To

This appendix develops, at implementation depth, the estimation layer that turns per-detection radar velocities into clubhead motion: what the current OpenFlight hardware can and cannot recover, why the twist (screw) representation of rigid-body motion is the natural formalism for the problem, and a step-by-step recipe for building the estimator on the IWR6843-class sensor described in Chapter C. It extends the EKF architecture of Section B.2 from point tracking (the ball) to rigid-body tracking (the club).

E.1 Context: what the commercial systems actually recover

TrackMan's club-data pipeline, as described in its documentation and patent family, tracks the clubhead "from about knee height to impact," resolves the head's radar return into velocity components across multiple receivers, and reports the trajectory of the head's *geometric center* [67, 72, 66]. The physical basis is that a rotating, translating clubhead is not a point target: the toe moves up to $\sim$7 mph faster than the heel, so the club's return occupies a spread of Doppler bins, and multi-receiver phase interferometry (Section 4.2) locates each velocity component in angle. What is fitted to those observations is a *rigid-body motion model*, not an image; "3D silhouette" is marketing shorthand for that model fit.

Is screw theory used by the industry? There is no public evidence that TrackMan (or any launch-monitor vendor) formulates this fit in screw/twist coordinates: the patents describe point-trajectory tracking, Doppler-component decomposition, and silhouette correlation, without reference to the screw formalism [71, 72]. The instantaneous screw axis (ISA) *is*, however, an established tool in golf biomechanics: Vena et al. applied ISA theory to optical motion capture of the swing in a two-part *Sports Engineering* study, verifying that segment motion during the downswing is dominantly rotational about a well-defined moving axis (at least 71% of marker velocity attributable to ISA rotation) and using ISA smoothness to characterize the kinematic sequence [77, 78]. The contribution proposed here — estimating the club's twist *directly from Doppler detections* rather than from marker positions — is, to our knowledge, not published in the golf literature, although the underlying mathematics is standard in robotics [34] and radar micro-Doppler analysis [5]. For OpenFlight this is an opportunity: the formalism is public-domain mathematics, distinct from the specific claimed pipelines in Chapter D.

E.2 Screw theory in the minimum required dose

A rigid body's instantaneous motion is fully described by a **twist**

$$\xi = (\boldsymbol{\omega},\, \boldsymbol{v}_O) \in \mathfrak{se}(3), \tag{E.1}$$

where $\boldsymbol{\omega}$ is the angular velocity and $\boldsymbol{v}_O$ is the linear velocity of a chosen body reference point O. The velocity of any body-fixed point at position $\boldsymbol{r}$ relative to O is then

$$\boldsymbol{v}(\boldsymbol{r}) = \boldsymbol{v}_O + \boldsymbol{\omega} \times \boldsymbol{r}. \tag{E.2}$$

Chasles' theorem states that every such motion is instantaneously a rotation about, plus a translation along, a unique line in space: the **instantaneous screw axis** (ISA). Its direction is $\hat{\boldsymbol{\omega}}$; a point on it is

$$\boldsymbol{r}_{\text{ISA}} = \frac{\boldsymbol{\omega} \times \boldsymbol{v}_O}{\|\boldsymbol{\omega}\|^2}, \qquad h = \frac{\boldsymbol{\omega} \cdot \boldsymbol{v}_O}{\|\boldsymbol{\omega}\|^2}, \tag{E.3}$$

where the **pitch** h is the translation per radian along the axis. For a downswing near impact the club's motion is close to a pure rotation about a hub near the hands: the ISA passes near the grip and the pitch is small. That geometric fact is both a physical insight (the ISA trajectory *is* a rigorous definition of swing plane) and, later, a regularization prior. Standard references: Murray, Li & Sastry for the mathematics [34]; Vena et al. for golf-specific ISA practice and its error behavior [77].

E.3 The measurement model: Doppler is linear in the twist

A radar detection assigns to some scattering center at known position $\boldsymbol{p}_i$ (from range and monopulse angle) a radial velocity $\dot{d}_i$ along the unit line of sight $\hat{\boldsymbol{u}}_i = \boldsymbol{p}_i/\|\boldsymbol{p}_i\|$. Substituting Equation (E.2) with $\boldsymbol{r}_i = \boldsymbol{p}_i - \boldsymbol{p}_O$:

$$\dot{d}_i = \hat{\boldsymbol{u}}_i \cdot \boldsymbol{v}(\boldsymbol{r}_i) = \underbrace{\hat{\boldsymbol{u}}_i}_{1\times 3} \cdot \boldsymbol{v}_O \;+\; \underbrace{(\boldsymbol{r}_i \times \hat{\boldsymbol{u}}_i)}_{1\times 3} \cdot \boldsymbol{\omega}. \tag{E.4}$$

This is the reciprocal product of the sight line (as a Plücker line) with the twist, and it is **exactly linear** in the six unknowns $(\boldsymbol{v}_O, \boldsymbol{\omega})$. Each detection contributes one row of a linear system

$$\boldsymbol{z} = A\,\xi + \boldsymbol{\epsilon}, \qquad A_i = [\;\hat{\boldsymbol{u}}_i^{\mathsf{T}} \mid (\boldsymbol{r}_i \times \hat{\boldsymbol{u}}_i)^{\mathsf{T}}\;], \tag{E.5}$$

so the per-frame twist estimate is weighted least squares — no iteration, no linearization error, and a covariance $(A^{\mathsf{T}}WA)^{-1}\sigma^2$ for free. This is the property that makes the screw formulation not merely elegant but *operationally* correct for radar: the sensor's native observable is already a linear functional of the twist. (Camera systems enter the same framework from the other side: a fiducial-tracked face gives an $SE(3)$ pose per frame, and the matrix logarithm of the frame-to-frame relative pose is a finite twist [34]; one downstream representation serves both modalities.)

E.4 What the current hardware can observe

Apply Equation (E.5) to each OpenFlight sensor to see the ceiling precisely.

E.4.1 OPS243-A: one row, no position

The OPS243-A has a single receive channel: it measures $\dot{d}_i$ but neither range nor angle, so $\boldsymbol{p}_i$ is unknown and Equation (E.5) cannot be assembled. What survives is the *marginal distribution* of $\dot{d}$ over the head: at 30 ksps with 128-sample segments the raw Doppler bin is ≈3.3 mph (Chapter C), so the full toe–heel spread of a driver spans only ∼2 raw bins. Software can and should still extract (Section B.1): the spread's midpoint or a fixed percentile as a stable club-speed reference (the peak is toe glint); the spread *width* as a crude proxy for $\|\boldsymbol{\omega}\|$ projected on the line of sight; and head/shaft separation by gating the slow tail. That is the honest ceiling of the current stack: a one-dimensional shadow of the twist, useful for de-biasing club speed, incapable of yielding path or attack angle.

E.4.2 K-LD7: the right equation, the wrong operating envelope

The K-LD7's dual receive patches provide per-bin monopulse angle — rows of Equation (E.5) in principle — but its fastest setting has a 100 km/h (≈62 mph) unambiguous-velocity ceiling and ∼29 ms frames (Chapter C), so a driver head aliases and traverses more than a meter between frames. It remains a ball-burst angle sensor, not a club tracker.

E.4.3 IWR6843: all three ingredients in one package

The IWR6843 supplies everything Equation (E.5) needs, given custom chirp firmware (Chapter C): range at 3.75 cm resolution (separates head from shaft and arms), per-chirp Doppler with both span and resolution set by the chirp plan, and per-detection monopulse angle from 12 virtual antennas. Representative chirp budget for the club problem:

Choice	Value	Consequence	Governing relation
Chirp repetition T_c	25 µs	$v_{\max} = \lambda/4T_c \approx 50\,\mathrm{m/s}$ (112 mph)	unambiguous velocity
Chirps per frame N	128	$\Delta v = \lambda/2NT_c \approx 0.8\,\mathrm{m/s}$	velocity resolution
Frame time	3.2 ms	∼300 Hz frame rate	swing-resolved motion
Bandwidth B	4 GHz	$\Delta R = c/2B = 3.75\,\mathrm{cm}$	3–4 cells across the head

At $\lambda = 5\,\mathrm{mm}$ the toe–heel spread (∼3 m/s) covers ∼4 velocity cells — twice the OPS243-A's resolving power, with each cell additionally localized in range and angle. The out-of-box SDK point cloud at 10–20 Hz is useless here; the custom chirp configuration plus on-chip range-Doppler processing is the enabling (and largest) engineering task, as flagged in Chapter C.

E.5 Estimation recipe, step by step

The following is the full pipeline, in execution order, for one swing. Notation: frames indexed k, detections within a frame indexed i.

Step 1 — Detect and localize

Run 2D CFAR on each frame's range-Doppler map (OS-CFAR preferred over cell-averaging: club and ball are exactly the closely-spaced-targets case it is designed for, Chapter C). For each detection, compute monopulse azimuth/elevation from the virtual-array phase and form $\boldsymbol{p}_i = (R_i, \theta_i, \phi_i)$ in Cartesian sensor coordinates, with a per-detection covariance driven by SNR (angle variance $\propto 1/\mathrm{SNR}$; range variance from the window function).

Step 2 — Segment the club

Gate detections to the club by a joint predicate: range inside the swing corridor, radial speed in the club band (above waggle, below ball speeds), and — once tracking has started — Mahalanobis distance to the propagated track. Reject the shaft by its distinct signature: shaft returns sit at lower speed and nearer range, forming an elongated cluster whose major axis points at the head cluster. Body and arm clutter falls out on speed alone.

Step 3 — Per-frame twist solve

Choose the body reference point O_k as the SNR-weighted centroid of the head cluster (this choice only re-parameterizes $\boldsymbol{v}_O$; the twist itself is frame-invariant). Assemble Equation (E.5) over the frame's m_k detections and solve weighted least squares with a robust loss (Huber or Cauchy) on the residuals — robustness is not optional, because *glint migration* (specular points wandering across the curved metal head as aspect changes) violates the fixed-scatterer assumption at the few-centimeter level and manifests as heavy-tailed residuals. Record $\hat{\xi}_k$ and its covariance $\Sigma_k = (A^\mathsf{T} W A)^{-1}$.

Step 4 — Read the conditioning honestly

Compute the SVD of the weighted A. Expect, from a single vantage $\sim$2 m away with detections spanning $\sim$10 cm: one strong singular value (bulk radial velocity — essentially club speed), one or two moderate ones (the transverse velocity component resolved by angle diversity, and the $\boldsymbol{\omega}$ combination that generates the Doppler spread), and the rest weak. **Do not invert the weak directions per frame.** Either truncate the SVD (solve only the well-conditioned subspace and carry the null-space explicitly) or defer those components entirely to the smoother of Step 5. A per-frame 6-DOF solve that "works" numerically will hallucinate the unobservable components from noise; this is the single most likely implementation failure mode.

Step 5 — Smooth the twist trajectory on $SE(3)$

The rank deficiency resolves *across* frames: as the head sweeps through the beam over $\sim$30 ms ($\sim$10 frames at 300 Hz), the sight-line geometry rotates and different frames constrain different twist combinations. Estimate a smooth trajectory rather than independent snapshots: state $x_k = (T_k \in SE(3),\, \xi_k)$, i.e., pose and twist; process model $T_{k+1} = T_k \exp\big((\xi_k \Delta t)^\wedge\big)$ with a slowly varying twist (white-noise angular/linear acceleration, tuned to swing dynamics — the head gains roughly 1 mph per millisecond late in the downswing); measurements = the raw rows of Equation (E.4) (preferable to feeding the Step-3 point estimates, since rows carry their true information content); run an extended/unscented filter with the standard Lie-group state handling [34], then an RTS backward smoother, exactly as in Section B.2. Add two physically motivated priors, weighted weakly enough to be overruled by data:

1. **Low pitch**: penalize $h^2 = (\boldsymbol{\omega} \cdot \boldsymbol{v}_O)^2 / \|\boldsymbol{\omega}\|^4$ — the downswing is close to a pure rotation [77].
2. **Hub proximity**: penalize distance of the ISA (Equation (E.3)) from a broad prior region around the golfer's hands.

Step 6 — Evaluate at impact, then stop

Rigidity holds before impact and fails during it ($\sim$500 µs of gross deformation), so fit only up to the last pre-impact frame and evaluate the smoothed twist at the impact timestamp (from the sound trigger, minus the acoustic delay budget of Chapter C). This mirrors the commercial convention: TrackMan defines club speed "just prior to first contact" [67].

Step 7 — Project out the parameters

All club-delivery parameters are now projections of one estimated object:

Parameter	Projection of the impact-time twist
Club speed	$\lVert \boldsymbol{v}_O + \boldsymbol{\omega} \times \boldsymbol{r}_{gc} \rVert$ at a *declared* reference point $\boldsymbol{r}_{gc}$ (cluster centroid, or a per-club calibrated offset)
Club path	horizontal direction angle of that velocity vector vs. the target line
Attack angle	vertical direction angle of the same vector
Closure rate	component of $\boldsymbol{\omega}$ about the estimated shaft axis, in °/s — the GCQuad-exclusive parameter (Section 5.3), free here
Swing plane / direction	orientation of the ISA (Equation (E.3)) and of the plane its trajectory sweeps
Low point	extrapolate the reference-point arc to its velocity-vertical zero
Gear-effect recoil	(post-impact, optional) the discontinuity in $\boldsymbol{\omega}$ across impact estimates the head's angular recoil, cross-checkable against Equation (3.8)

Report each with the covariance propagated from the smoother, and tag provenance per Table 2.3: with this pipeline, path and attack angle move from *derived* to *measured*, while face angle and impact location remain optics problems (Section 4.5.2).

E.6 Validation plan

Validate in order of increasing realism, reusing the standards of Chapter 9:

1. **Synthetic**: simulate scatterers on a CAD clubhead following a recorded swing trajectory, generate detections with realistic SNR and glint migration, and verify the recovered twist against truth — this exercises Steps 3–5 without hardware and quantifies the observability claims of Step 4.

2. **Pendulum rig**: a club swung as a physical pendulum has an analytically known, planar, fixed-axis twist (pitch exactly zero, ISA fixed); any bias in the pipeline shows up immediately.

3. **Cross-device**: compare club speed, path, and attack angle against the MLM2PRO per the protocol of Chapter 9, with the reference-point caveat of Section 4.5 stated up front; sanity-gate every shot against the loft-dependent smash ceiling (Section 3.2).

4. **Internal consistency**: the measured path must satisfy the swing-geometry identity (path as a function of swing direction, swing plane, attack angle — Chapter 3) within uncertainty; violations indicate reference-point drift or alignment error.

Implication for OpenFlight

The twist formulation costs nothing extra to adopt — the per-frame solve is a small weighted least squares — and buys three things the point-tracking alternative cannot: a principled, declared club-speed reference point (ending the largest inter-device disagreement), closure rate and a rigorous swing plane as free by-products, and honest covariances on every club parameter. It should be specified as the estimation layer for the IWR6843 integration from day one, with the OPS243-A velocity-distribution analytics (Section B.1) as the degenerate 1D special case of the same model.

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